

Green subsidies with demand distortions

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Abstract

Credit constraints and other demand distortions are pervasive in lower-income contexts. A model of clean technology adoption shows that demand distortions can increase the welfare gains from green subsidies. This arises through two channels: shifting the marginal adopter toward higher private and social benefits, and increasing demand elasticity. We test this by cross-randomizing purchase price subsidies, usage subsidies, and access to an interest-free loan for a US\$84 induction stove among 2,134 charcoal users in Kenya. Usage subsidies of 75% have a precise zero effect on adoption and usage. Purchase price subsidies, however, generate large welfare gains. Credit constraints raise the subsidy's efficacy by 35%–73%, up to US\$10–US\$12 in observable welfare gain per subsidy dollar. This demand distortion also lowers the subsidy cost of abatement by 41%, to US\$12 per tCO₂e. This operates through the two channels identified in the theory: credit constraints raise the marginal adopter's positive externality by 19% and fuel savings by 30%, and lower the subsidy cost per marginal adoption by 29%. In a counterfactual without demand distortions, the abatement cost rises to US\$137 per tCO₂e. Contexts with large demand distortions could offer many cost-effective opportunities for climate change mitigation financing.

JEL: H23, Q5, O13

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1 Introduction

Climate change mitigation technologies often generate social benefits only if an initial investment is followed by high usage: for example, an electric vehicle abates emissions only when driven in place of a gasoline vehicle. When usage generates private benefits like fuel savings, the market selects well: agents with the largest social benefits also have the highest willingness-to-pay (WTP) and adopt first.¹ However, this same force creates an adverse selection problem for adoption subsidies. Agents with large private benefits already adopt, so on the margin the subsidy induces agents with lower usage who generate less social benefit. Furthermore, subsidy expenditures accrue disproportionately to inframarginal agents, whose behavior is unchanged. These dynamics have dampened the social benefits of a wide range of programs subsidizing upfront investment.² This has led many economists to favor a marginal-cost subsidy, which might avoid these problems: by aligning private and social benefits, it draws in adopters in proportion to the abatement they generate and incentivizes usage on the intensive margin (Pigou, 1920).

This logic assumes WTP reflects private benefit. However, in many settings demand distortions drive WTP below the private benefit (Atkin et al., 2025).³ Capital market failures prevent agents from accessing future payment streams to fund present-day investments (de Mel et al., 2008; Banerjee and Duflo, 2014). Behavioral biases can cause agents to undervalue benefits (Laibson, 1997; DellaVigna, 2009; Duflo et al., 2011). Transaction costs and savings constraints can distort household energy purchasing (Abubakari et al., 2026). As a result, even agents with large private benefits may not make the initial investment, leaving them marginal to a subsidy. Many such distortions result from underlying institutional constraints and market failures that cannot be resolved cost-effectively with policy interventions.

This paper’s main research finding is that, relative to an undistorted market, distortions can make green subsidies more effective through two channels: by changing the marginal adopter’s benefit, and by changing the subsidy cost per marginal adopter. At the same time, some distortions generate economic costs, undermining the subsidies’ welfare gains. We first develop a theory of subsidy targeting with demand distortions and then evaluate it empirically in the context of induction cookstove adoption in Kenya. This setting exhibits severe capital market distortions: buyers paying in installments normally face an annualized percentage rate (APR) of 224%. We implement a randomized trial among 2,134 charcoal-cooking households in which we (i) vary fixed-cost (purchase price) and marginal-cost (usage) subsidies; (ii) manipulate demand distortions by offering interest-free loans; and (iii) causally estimate the private and social benefits of a green technology. We find that credit constraints increase the marginal externality of a purchase subsidy and decrease its cost per induced adoption. Together, these effects increase subsidy welfare gains by 35%–73%.

¹When follow-through is costly (e.g., forest conservation, sustainable agriculture), agents with low follow-through invest first. Incentivizing usage requires costly information and enforcement capacity, and may introduce moral hazard (Oliva et al., 2020; Jack et al., 2025; Aspelund and Russo, 2026; Chen et al., 2026).

²See for example Boomhower and Davis (2014), Rapson and Muehlegger (2023), and Hahn et al. (2026).

³Some papers refer to the observable object as ability-to-pay (e.g., Banerjee, 1997). We, and others, use WTP to denote observed demand (e.g., Cohen and Dupas, 2010; Tarozzi et al., 2014; Burlig et al., 2026).

Usage subsidies, on the other hand, have no detectable impact on WTP or usage.

The model evaluates the use of purchase price subsidies and usage subsidies under two types of distortions: allocation distortions, which only affect decision-making (such as inattention and present bias), and resource distortions, which also incur resource cost and thus affect both decision-making and payoffs (such as transaction costs and credit constraints). We offer both a general framework and a specialization to our context, which is characterized by positively correlated private and social benefit, a resource distortion, and inelastic energy demand. The model yields two results: subsidies generate larger welfare gains in contexts with larger demand distortions, and purchase price subsidies can dominate usage subsidies in such contexts. We characterize conditions under which these results hold even if the distortion adds real economic costs.

To test the model’s predictions, we cross-randomize purchase price subsidies of either 13% or 76% off the US\$84 price of the induction stove bundle (which includes three pieces of compatible cookware) with usage subsidies that lower the price of electricity used to operate the stove by 0%, 25%, or 75% for six months (with no change in the cost of electricity for other uses). We randomize access to an interest-free, four-month loan designed to relax credit constraints, a key source of distorted demand in this context. We elicit WTP using an incentive-compatible Becker, DeGroot, and Marschak (BDM; 1964) mechanism. 29% of study participants bought the stove bundle, of whom 67% bought it with credit, with repayment averaging 86%. We convert fuel expenditures to tons of carbon dioxide equivalent (tCO_2e) using market prices and local deforestation rates.

The induction stove generates large private and social benefits. High-frequency temperature sensors record a reduction in daily charcoal cooking of 30 minutes, and self-reported monthly charcoal spending decreases by US\$9.7. Self-reported monthly electricity spending increases by US\$2.5, which aligns with the induction stove’s recorded usage at local tariffs. Aggregate changes in biomass, electricity, and liquefied petroleum gas (LPG) usage save US\$8.1 per month and abate 2.4 tCO_2e per year. We apply these annual estimates over the 2.2-year lifespan of the stove, estimated using historical usage data from the manufacturer.

We find that usage subsidies have minimal impacts, if any. While we provided detailed information and drew attention to cooking costs before eliciting WTP, each US\$1 in usage subsidy has a precisely estimated US\$0.0 impact on WTP (95% CI: $[-\text{US\$0.2}, \text{US\$0.2}]$), and we can rule out that a 1% usage price decrease increases electric stove usage by more than 0.3%. These results hold with and without access to credit.

Purchase price subsidies, however, generate large welfare gains. Each subsidy dollar generates US\$3.4 in fuel savings and US\$10 in environmental benefit when valuing CO_2e at a social cost of carbon (SCC) of US\$120 per ton (EPA, 2023). The subsidy abatement cost of US\$12 per tCO_2e is roughly one-tenth of the SCC. When factoring in the stove manufacturing cost and the resource costs caused by the distortion, we estimate that each subsidy dollar generates an observable welfare gain of US\$10–US\$12: the range reflects uncertainty in what generates the remaining under-adoption.⁴

⁴We define the observable welfare gain as the private fuel savings plus the environmental benefit valued at the accepted SCC, minus the manufacturing cost, and minus any resource cost.

The paper’s main empirical finding is that demand distortions substantially increase the welfare gains generated by these subsidies. Average WTP in the control group is only 10% of total discounted fuel savings: access to an interest-free loan increases this to 19% ($p\text{-val} < 0.001$). This demand distortion increases the observable welfare gain of purchase price subsidies from US\$7–US\$8 up to US\$10–US\$12 per dollar of government spending. It furthermore decreases the subsidy cost per tCO_{2e} from US\$20 to US\$12.

Distortions increase subsidy efficacy through the two channels identified in the theory. First, in the presence of larger credit distortions, adopters who are marginal to the subsidy generate 30% more in private fuel savings (US\$228 versus US\$175; $p\text{-val} = 0.058$) and abate 19% more CO_{2e} (5.7 tCO_{2e} versus 4.8 tCO_{2e}; $p\text{-val} = 0.048$). Second, credit constraints shift demand elasticity from -0.9 to -2.3 , reducing subsidy expenditure per marginal sale by 29%, from US\$96 to US\$67.

The random variation from the experiment allows us to estimate the model and generate counterfactual estimates of subsidy efficacy in contexts with smaller demand distortions. In a counterfactual without distortions, where WTP equals the private benefit, the subsidy cost would rise to US\$137 per ton of CO_{2e}. While the model extrapolates beyond what we observe in the experiment, abatement in an undistorted counterfactual is consistently at least ten times as expensive as in our context across a range of specifications and assumptions.

Global carbon mitigation financing exceeds US\$1 trillion annually (Buchner, 2024). Where should the marginal dollar go? An efficient allocation of global mitigation financing requires directing each marginal dollar towards the opportunities with the lowest marginal abatement costs, and then moving up along the abatement supply curve over time (Gillingham and Stock, 2018; Kolstad, 2011). Our results suggest that contexts with large demand distortions could contain many of the lowest-cost portions of the global abatement supply curve. In counterfactuals that hold technology fixed, demand distortions alone decrease the subsidy cost of abatement by an order of magnitude. This could help explain variation in subsidy costs across settings. For example, 80% of new cars sold in the U.S. are financed (Experian, 2023), reflecting a well-functioning consumer credit market, and U.S. electric vehicle subsidies cost over US\$1,200 in government expenditure per tCO_{2e} abated (Hahn et al., 2026)—100 times the US\$12 per ton in our setting. While the underlying technologies also differ, the counterfactual analyses suggest that demand distortions can account for a meaningful part of this gap.

Related literature This paper contributes to a growing literature characterizing how demand distortions—or ‘wedges’—affect market outcomes, including Atkin et al. (2025), Bergquist et al. (2026), Ghatak and Mookherjee (2025), and Indarte et al. (2026). This follows a large body of development economics research documenting that pervasive market failures dampen WTP to below private benefit, and that credit constraints in particular depress adoption of privately beneficial technologies.⁵ We study the implications of these distortions for environmental subsidies.

⁵Cohen and Dupas (2010), Tarozi et al. (2014), Burlig et al. (2026), Banerjee and Duflo (2014), de Mel et al. (2008), Fafchamps et al. (2014), McKenzie and Woodruff (2008), Duflo et al. (2008), Gertler et al. (2024), Adhvaryu et al. (2020), and Berkouwer and Dean (2022).

A well-established literature on selection and mechanism design for green subsidies evaluates how the relationship between agents’ private willingness to participate and the social value of their actions affects the welfare consequences of subsidy programs (Aspelund and Russo, 2026; Chen et al., 2026; Ito et al., 2023). A related strand of literature evaluates targeting of energy subsidies in the presence of internalities (e.g., Allcott and Taubinsky, 2015; Allcott and Greenstone, 2024; Gillingham and Palmer, 2014; Allcott et al., 2014). We connect these literatures with the large development economics literature on pervasive demand distortions in low- and middle-income countries. We document not just that distortions affect decision-making, but that they can affect the aggregate efficacy of green subsidies. Furthermore, the distortions prevalent in developing countries—like credit constraints and transaction costs—often entail real economic costs, which enter into the social planner’s welfare function. Notably, Abubakari et al. (2026) study the efficiency properties of subsidy instruments in the presence of demand distortions that incur resource costs. We build on this by randomizing the distortion itself.

We contribute to a longstanding literature on optimal environmental policy instruments and second-best environmental policies (Jacobsen et al., 2020; Knittel and Sandler, 2018; Holland et al., 2016; Fowlie et al., 2016; Allcott and Taubinsky, 2015). Existing work has documented that second-best instruments (like fixed-cost subsidies) can be more efficient than first-best, Pigouvian instruments that operate on the relevant margin (e.g., Carlton and Loury, 1980; De Groote and Verboven, 2019). We build on this by adding demand distortions as a broad class of second-best settings that can warrant the use of second-best environmental policies.

A large empirical energy economics literature, primarily in high-income countries, studies intensive and extensive margin demand elasticities (Ito and Zhang, 2025; Ito, 2014; Kahn and Wolak, 2013; Ida et al., 2026; Hahn and Metcalfe, 2021; Houde and Aldy, 2017; Boomhower and Davis, 2014; Borenstein and Davis, 2025). We estimate extensive and intensive margin responses in a very different context. We find a substantially more elastic extensive margin response to purchase price subsidies, but we detect no extensive margin or intensive margin response to usage subsidies. This contributes to the policy debate around electricity tariffs designed to spur decarbonization in low- and middle-income countries. Our research design circumvents a common limitation of usage subsidies, which can sometimes inadvertently incentivize the use of electricity for other purposes (Borenstein, 2025; Campbell, 2025).

Finally, we contribute to a small but growing literature on electric cooking. More than two billion people lack modern cooking technologies (IEA, 2024). With billions gaining access to electricity in recent decades, widespread electric cooking has become an achievable policy goal (World Bank, 2024). Using experimental variation, Desbureaux et al. (2025) and Mahadevan et al. (2025) find that purchase price subsidies increase adoption of electric cooking technologies in the Democratic Republic of the Congo (DRC) and in Nepal, respectively, and generate important social benefits. Gould et al. (2023) find that the nationwide transition from gas towards electric cooking reduced Ecuador’s CO_{2e} emissions. Other descriptive research and laboratory experiments have shed light on perceptions, usage patterns, energy efficiency, and potential for scale of electric cooking technologies

(Aemro et al., 2021; Leary et al., 2021; Saha et al., 2021; Clements et al., 2020; Alem et al., 2014; Banerjee et al., 2016; Rubinstein et al., 2022).

2 Theory

We evaluate a green technology whose purchase enables usage, and whose sustained use generates a positive externality. A principal can offer either a usage subsidy, which lowers the marginal cost of usage, or a purchase price subsidy, which lowers the cost of buying the technology.

The model evaluates how a demand distortion can change the return to a purchase price subsidy, and in what settings a purchase price subsidy can generate larger welfare gains than a usage subsidy. We first isolate two channels through which distortions can affect marginal and average returns in a generalized framework. We then apply a specialized version of the model to our study setting and compare the returns to purchase price and usage subsidies.

While Pigouvian taxes are theoretically first-best (Pigou, 1920), we focus on green subsidies for several reasons. Carbon taxes are often politically intractable (consider the repeal of Canada’s well-designed carbon tax; Department of Finance Canada, 2025). Low-income households spend a disproportionate share of their incomes on energy costs: Kenya recently froze electricity tariff reviews amid “public discontent over fuel prices” (Daily Nation, 2026). Taxation may also be logistically infeasible (firewood and charcoal production is often informal). Finally, many environmental principals—such as multilateral donors, environmental agencies, the United Nations Framework Convention on Climate Change (UNFCCC), carbon offset markets, philanthropists, and investors—lack the legal authority to levy a tax and therefore rely exclusively on subsidies to spur green technology adoption.

2.1 A generalized framework

Consider a mass of agents indexed by their undistorted willingness-to-pay (WTP) v for adopting a green technology, with continuous distribution F , density f , and survival function $Q(v) \equiv 1 - F(v)$. Each adoption generates an external benefit; let $\Phi(v)$ denote its conditional mean among agents with undistorted WTP v .

A demand distortion maps the undistorted WTP v into the decision-relevant WTP $D(v)$.⁶ We assume the distortion can only dampen WTP, $D(v) \leq v$. Let p be the fixed competitive market price, let s_1 be a fully passed-through purchase price subsidy, and let $P = p - s_1$ denote the net purchase price faced by the buyer. An agent adopts when $D(v) \geq P$. Write $\tilde{Q}(P) = \mathbb{P}(D(v) \geq P)$ for adoption under the distortion.

In some cases the distortion affects decision-making but does not affect payoffs conditional on the actions taken. In other cases the distortion also consumes real resources. We refer to these as allocation distortions and resource distortions, respectively. Let $\rho(P)$ denote the resource cost per

⁶We assume that D is continuously differentiable with $D'(v) > 0$ over the relevant range and that all cutoffs are interior, with positive density and positive remaining demand.

Table 1: Examples of distortions that lower willingness-to-pay

Distortion	$D(v)$	$\rho(P)$	Examples
<i>Distortions that may consume resources</i>			
Credit markup	$v/(1+r)$	rP	<i>This paper</i>
Hassle, transaction costs	$v - c$	c	Abubakari et al. (2026)
<i>Pure allocation distortions ($\rho = 0$)</i>			
Present bias, inattention	$(1 - \theta)v$	–	Duflo et al. (2011); Allcott and Wozny (2014)
Additive bias	$v - b$	–	Allcott and Taubinsky (2015)
Ironing	$\mathbb{E}[w \mid w \leq v]$	–	Ito (2014); Rees-Jones and Taubinsky (2020)
Convex misperception	$v^2/\bar{v}$	–	Bushong et al. (2021)

Notes: Examples of increasing mappings that lie weakly below the identity and are strictly below it over the relevant interior. The parameters satisfy $r, c, b > 0$ and $\theta \in (0, 1)$. w denotes the value of an inframarginal unit; $\bar{v}$ denotes the maximum willingness-to-pay in the support. Allocation distortions have $\rho = 0$. For resource distortions, ρ is shown assuming the principal fully internalizes the resource costs.

adopter internalized by the principal. It is zero when either the distortion is an allocation distortion or the principal does not care about the private costs and benefits of adopters (e.g., some principals might only value emissions reductions). The remainder of this subsection assumes the principal values the resource cost as a social planner would.

Table 1 gives examples of distortions that fit this representation. Although their underlying mechanisms differ, each can be summarized as a mapping from undistorted WTP v to distorted WTP $D(v)$.

2.1.1 Channel 1: Selection

For any given price P , the undistorted and distorted marginal adopters, $\underline{v}$ and $\tilde{v}$, satisfy

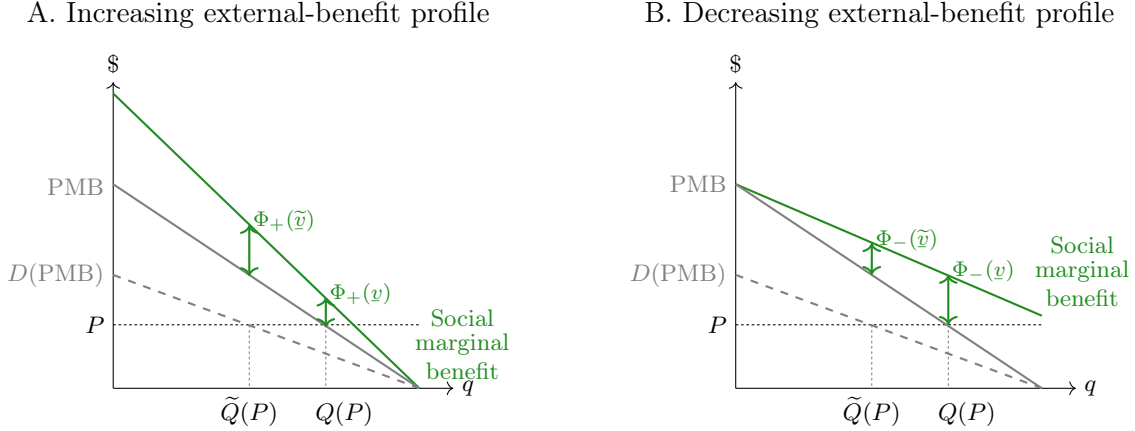
$$\underline{v} = P = D(\tilde{v}) \leq \tilde{v}.$$

The distortion weakly reduces adoption and moves the margin toward types with higher private values. Whether this raises or lowers the marginal externality depends on the slope of $\Phi(v)$. If $\Phi(v)$ is increasing over the relevant range, then the distortion increases the marginal adopter's externality ($\Phi(\tilde{v}) \geq \Phi(\underline{v})$). This reverses if $\Phi(v)$ is decreasing. Figure 1 illustrates this.

2.1.2 Channel 2: Demand elasticity

The demand elasticity determines the number of additional adoptions induced by each purchase price subsidy dollar. A marginal increase ds_1 in the purchase price subsidy induces $-Q'(P) ds_1$ new adoptions and raises expenditure on existing adopters by $Q(P) ds_1$. Let κ denote the inframarginal

Figure 1: How the external-benefit profile determines the role of selection



Notes: Social marginal benefit (SMB) equals private marginal benefit (PMB) plus the conditional external benefit. The distortion increases the gap when $\Phi(v)$ is increasing ($\Phi_+(\tilde{v}) > \Phi_+(v)$) and decreases it when $\Phi(v)$ is decreasing ($\Phi_-(\tilde{v}) < \Phi_-(v)$).

expenditure per induced adoption. Let $|\mathcal{E}(P)| \equiv -PQ'(P)/Q(P)$ denote the price elasticity. Then:

$$\frac{\tilde{\kappa}(P)}{\kappa(P)} = \frac{\tilde{Q}(P) / \tilde{Q}'(P)}{Q(P) / Q'(P)} = \frac{|\mathcal{E}(P)|}{|\tilde{\mathcal{E}}(P)|}.$$

Thus, if $|\mathcal{E}(P)| < |\tilde{\mathcal{E}}(P)|$ then $\kappa(P) > \tilde{\kappa}(P)$: a distortion that raises elasticity lowers inframarginal subsidy expenditure per induced adoption in inverse proportion, increasing the number of additional adoptions per subsidy dollar.

2.1.3 Return to a purchase price subsidy

Let $\mathcal{V}(v)$ denote the contribution of adoption by type v to the principal's objective, excluding any resource cost. For a social-welfare objective it can include private and social benefits net of the technology's real economic cost. For a principal with only environmental objectives it may include only the externality. Let W denote the aggregate objective, and let C denote total subsidy expenditure. A marginal increase in s_1 induces agents at $\tilde{v} = D^{-1}(P)$. The principal spends $s_1 + \tilde{\kappa}(P)$ per induced adoption: s_1 on the new adopter and $\tilde{\kappa}(P)$ on existing adopters.

Aggregate resource cost $R(s_1)$ comprises the per-adopter resource costs $\rho(P)$ of inframarginal and induced adoptions. A subsidy adds resource cost among induced adopters but lowers the resource cost among inframarginal adopters when $\rho' > 0$:

$$R(s_1) \equiv \rho(p - s_1)\tilde{Q}(p - s_1), \quad \dot{R}(s_1) = \frac{R'(s_1)}{\tilde{Q}'(s_1)} = \underbrace{\rho(p - s_1)}_{\text{induced adopters' resource cost}} - \underbrace{\rho'(p - s_1)\tilde{\kappa}(p - s_1)}_{\text{inframarginal adopters' resource savings}}. \quad (1)$$

Proof in Online Appendix VI.1

Taken together, the principal's welfare return to an additional purchase price subsidy dollar is

$$\frac{dW/ds_1}{dC/ds_1} = \frac{\mathcal{V}(\tilde{v}) - \dot{R}(s_1)}{s_1 + \tilde{\kappa}(P)}. \quad (2)$$

Proof in Online Appendix VI.1

For a finite subsidy relative to no subsidy,⁷ let $\Delta Q \equiv \tilde{Q}(p - s_1) - \tilde{Q}(p)$ denote the number of adoptions induced by the subsidy and $\zeta \equiv \tilde{Q}(p - s_1)/\Delta Q$ the number of subsidy recipients per induced adoption. Let $\bar{\mathcal{V}}$ denote the average value of an induced adoption to the principal and ΔR the change in aggregate resource cost. Let $\Delta C = s_1 \tilde{Q}(p - s_1)$ denote total subsidy expenditure. Then average welfare generated per subsidy dollar is

$$\frac{\Delta W}{\Delta C} = \frac{\bar{\mathcal{V}} - \Delta R/\Delta Q}{s_1 \zeta}. \quad (3)$$

Proof in Online Appendix VI.1

The marginal and average cases have the same structure: selection determines the value of induced adopters, elasticity determines expenditure per induced adoption, and resource costs enter separately.

2.1.4 Return to a usage subsidy

We now allow the principal to subsidize usage at a per-unit rate $s_2 \geq 0$ and evaluate both instruments at the first dollar, $s_1 = s_2 = 0$. Let $x(v)$ denote usage conditional on adoption in the absence of a subsidy, with $x'(v) > 0$. The usage subsidy raises type v 's undistorted WTP by $b(v, s_2)$, with $b(v, 0) = 0$. The distortion applies to total WTP, so type v adopts when $D(v + b(v, s_2)) \geq P$; by the envelope theorem $\partial b/\partial s_2 = x(v)$ at $s_2 = 0$. A marginal usage subsidy therefore moves the adoption margin through the marginal adopter's usage $x(\tilde{v})$, but its cost reflects average usage over all adopters, $\bar{x}$.⁸ Induced adoptions per subsidy dollar are therefore

$$\left. \frac{d\tilde{Q}/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = \frac{1}{D'(\tilde{v})} \frac{f(\tilde{v})}{Q(\tilde{v})}, \quad \left. \frac{d\tilde{Q}/ds_2}{dC/ds_2} \right|_{s_1=s_2=0} = \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})}. \quad (4)$$

Proof in Online Appendix VI.2

The usage subsidy's advantage is that it can change usage among existing adopters. Let $\mathcal{V}_2(v, 0)$ denote the marginal effect of s_2 on the principal's value from type v at $s_2 = 0$, and define

$$\mathcal{I}_2 \equiv \frac{\int_{\tilde{v}}^{\infty} \mathcal{V}_2(v, 0) dF(v)}{\tilde{Q}(p) \bar{x}}, \quad (5)$$

⁷ Assume $0 < s_1 < p$ and a positive mass of induced adopters, so that $\Delta Q > 0$.

⁸ D' does not appear in the usage-subsidy expression in Eq. 4 because a distortion that dampens the usage subsidy's pass-through into WTP also compresses agents along that scale, so the two effects cancel.

the value generated per subsidy dollar by this intensive-margin effect. A marginal increase in s_2 increases spending on inframarginal adopters by $\tilde{Q}(p)\bar{x}$, induces $f(\tilde{v})x(\tilde{v})$ more adopters on the extensive margin, and generates $\mathcal{I}_2$ per dollar on the intensive margin. Let $R(s_1, s_2)$ denote aggregate resource cost, with $R(s_1, 0) = R(s_1)$, and let R_1 and R_2 denote its partial derivatives at $s_1 = s_2 = 0$. The first-dollar return to the usage subsidy is then

$$\left. \frac{dW/ds_2}{dC/ds_2} \right|_{s_1=s_2=0} = \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}) + \mathcal{I}_2 - \frac{R_2}{\tilde{Q}(p)\bar{x}}. \quad (6)$$

Proof in Online Appendix VI.2

Evaluating the return of the previous section at $s_1 = 0$ gives the analogous expression

$$\left. \frac{dW/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = \frac{1}{D'(\tilde{v})} \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}) - \frac{R_1}{\tilde{Q}(p)}. \quad (7)$$

Proof in Online Appendix VI.2

On the extensive margin, each dollar of purchase price subsidy lowers the adoption threshold by $1/D'(\tilde{v}) \geq 1$ dollars of undistorted WTP whenever $D'(\tilde{v}) \leq 1$. Since usage-subsidy dollars are paid on the usage of all adopters, only the share $x(\tilde{v})/\bar{x} \leq 1$ lowers the adoption threshold. The purchase price subsidy therefore induces weakly more adoptions per dollar whenever $D'(\tilde{v}) \leq 1$.

On the intensive margin, the usage subsidy changes usage among existing adopters, generating value $\mathcal{I}_2$ per dollar. The usage subsidy therefore generates a higher return if and only if its intensive-margin advantage, net of any differences in resource cost ($R_2/(\tilde{Q}(p)\bar{x}) - R_1/\tilde{Q}(p)$), exceeds the purchase price subsidy's extensive-margin advantage.

Proof in Online Appendix VI.2

2.2 Credit constraints and energy-efficiency adoption

We now specialize the framework to the distortion studied in the empirical application, which is a credit market failure. In our setting, institutional constraints prevent the establishment of policy responses such as credit bureaus, deposit insurance, and collateral posting and repossession that could otherwise help overcome credit market failures like adverse selection and moral hazard (Acemoglu et al., 2005; Bernanke, 2023). We consider a principal who fully internalizes private benefits and resource costs.

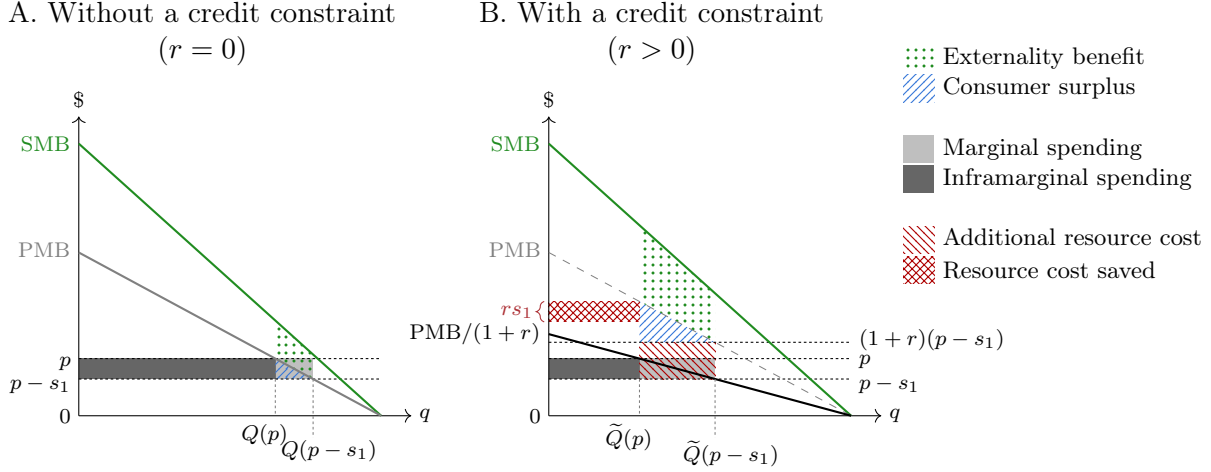
First, we characterize the credit market failure as a high interest rate $r > 0$, which raises the cost of the purchase to $(1+r)P$. The high r results from resources such as repayment monitoring and follow-up enforcement needed to offer loans in this context, given its market failures. The agent adopts when

$$v \geq (1+r)P, \quad \text{or equivalently} \quad D(v) = \frac{v}{1+r} \geq P.$$

Aggregate distorted demand is given by $\tilde{Q}(P) = Q((1+r)P)$. Since distorted elasticity at P equals undistorted elasticity at $(1+r)P$, the distortion raises elasticity whenever $|\mathcal{E}|$ is increasing in price; an increasing hazard rate f/Q is sufficient.

Proof in Online Appendix VI.3

Figure 2: Private value, social value, resource cost, and expenditure under a purchase price subsidy



Notes: Private marginal benefit (PMB) and social marginal benefit (SMB) in the context of a credit market distortion. The dotted green area is the externality generated by induced adopters, the striped blue area is their consumer surplus, and the gray areas are subsidy expenditure. In Panel B, the striped red area is the resource cost incurred by induced adopters. The crosshatched area is the resource cost saved when the subsidy reduces the amount financed for inframarginal adopters. The gray areas measure government expenditure; they are not themselves resource costs.

Second, we assume that fuel savings and emissions reductions both scale with clean technology usage. The correlation between emissions reductions and fuel savings varies with emissions factors and fuel prices, but in our data the correlation has an R^2 of 0.6.

Third, we model total cooking services as being perfectly inelastic: the usage decision is which technology to use for what proportion of total cooking time. In our data, we detect no change in the number of meals cooked per day (95% CI of [-0.18, 0.05] on a control mean of 2.8) after adoption. The 29-minute-per-day decrease in charcoal cooking time is accompanied by a 35-minute increase in electric cooking time with very little simultaneous stove use (Section 4.4 discusses this in more detail).

Figure 2 summarizes how a credit constraint changes the effects of a purchase price subsidy, using a pared-down version of the model to build graphical intuition. Agent i 's clean-technology usage after adoption is x_i (in the absence of a usage subsidy, which we return to later). Each unit displaces the incumbent technology by an amount that generates an external benefit ϕ and saves π in fuel costs, with fuel sold in a competitive market. Thus agent i 's private value is $v_i = \pi x_i$ (so that $v'(x) > 0$), and the external benefit of their adoption is $\Phi(v_i) = \phi x_i$.

The distortion increases subsidy efficacy in this context through two channels. Through the selection channel, the demand distortion causes a purchase price subsidy s_1 to generate a larger marginal positive externality. Through the elasticity channel, it lowers the subsidy spending required per additional adoption, as long as $f(v)/Q(v)$ is increasing. Induced adopters also incur an additional resource cost $r(p - s_1)$, but inframarginal adopters' resource cost falls by rs_1 .

2.2.1 Welfare effects of a purchase price subsidy

A purchase price subsidy s_1 generates $r(p - s_1)\Delta Q$ in resource cost among induced adopters (the striped red area in Panel B) but saves $rs_1\tilde{Q}(p)$ among inframarginal adopters (the crosshatched area).⁹ Applying this to Eq. 3, the average welfare gain per subsidy dollar is given by¹⁰

$$\mathcal{W}(r) \equiv \frac{\Delta W}{\Delta C} = \frac{\bar{V}(r) - rp}{s_1\zeta(r)} + r. \quad (8)$$

Proof in Online Appendix VI.3

The return is positive if and only if the additional resource cost for induced adopters does not outweigh the other benefits: $\bar{V}(r) + rs_1\zeta(r) > rp$.

The marginal effect of the interest rate on welfare per subsidy dollar is:

$$\mathcal{W}'(r) = \frac{1}{s_1\zeta(r)} \left[\underbrace{\bar{V}'(r)}_{\text{selection benefit}} - \underbrace{[\bar{V}(r) - rp]\frac{\zeta'(r)}{\zeta(r)}}_{\text{elasticity effect}} + \underbrace{s_1(\zeta(r) - 1)}_{\text{resource savings on inframarginal recipients}} - \underbrace{(p - s_1)}_{\text{resource cost of the induced purchase}} \right] \quad (9)$$

Proof in Online Appendix VI.3

When social value is increasing in undistorted WTP over the relevant range, a larger distortion raises $\bar{V}$ by moving the induced group toward higher-value agents (the selection channel). When $f(v)/Q(v)$ is increasing, an increase in r weakly lowers $\zeta(r)$ by increasing the share of recipients induced by the subsidy, which reduces expenditure per induced adoption (the elasticity channel). This increases welfare as long as the average value of an induced adoption exceeds the resource cost per adoption, $\bar{V}(r) > rp$.¹¹

Proof in Online Appendix VI.3

The resource savings effect arises because a larger distortion means each subsidy dollar paid to inframarginal adopters saves more financing resources. Working in the opposite direction, a larger distortion raises the resource cost of each induced purchase.

Taken together, welfare per subsidy dollar is locally increasing in r as long as the selection channel, the elasticity channel, and the resource savings together outweigh the additional resource cost.

⁹This assumes buyers finance the full net purchase price. If only part of the distortion is a resource cost, the same accounting applies to that part.

¹⁰We assume that the relevant objects are differentiable and that the cutoffs remain interior in a neighborhood of r ; primes denote derivatives with respect to r .

¹¹This may not hold if the primary welfare benefit of the subsidy is in reducing resource costs incurred by inframarginal households. In that case, the principal may want a subsidy that induces inefficient adoptions on the margin because the financing costs saved on inframarginal purchases outweigh the losses from those adoptions.

2.2.2 When can a purchase price subsidy outperform a usage subsidy?

Using $1/D' = 1 + r$ and $x'(v) > 0$ in Eq. 4, induced adoptions per subsidy dollar are given by

$$\left. \frac{d\tilde{Q}/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = (1+r) \frac{f(\tilde{v})}{Q(\tilde{v})} \geq \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})} = \left. \frac{d\tilde{Q}/ds_2}{dC/ds_2} \right|_{s_1=s_2=0}. \quad (10)$$

Proof in Online Appendix VI.3

We use Eq. 6 and Eq. 7 to compare usage and purchase price subsidies in the context of credit constraints. The purchase price subsidy has the larger first-dollar welfare return if and only if

$$[\mathcal{V}(\tilde{v}) - rp] \left[(1+r) - \frac{x(\tilde{v})}{\bar{x}} \right] \frac{f(\tilde{v})}{Q(\tilde{v})} + r > \mathcal{I}_2. \quad (11)$$

Proof in Online Appendix VI.3

The first term is the value of the purchase price subsidy's additional induced adoptions, net of their financing resource cost; it favors the purchase price subsidy when $\mathcal{V}(\tilde{v}) > rp$. The term r is the financing resource saving on existing purchases. The purchase price subsidy generates a larger return than the usage subsidy when these extensive-margin and resource-saving gains exceed the usage subsidy's intensive-margin return. A sufficiently large $\mathcal{I}_2$ reverses the ranking.

2.3 Model extensions

We now consider several model extensions that might affect these results; formal derivations are in Online Appendix VII.

Marginal cost of raising funds Introducing a positive marginal cost of raising funds would increase the relative marginal benefit of subsidies in a distorted market, since the distortion reduces inframarginal spending.

Taxation With a tax available, setting the marginal relative-price wedge equal to the externality can induce first-best usage (Pigou, 1920). Under a pure allocation distortion, only the adoption margin then remains distorted, so setting $s_1 = p - D(p)$ achieves the first-best allocation. However, under a resource distortion, no purchase price subsidy restores the first best because the distortion both affects the adoption decision and consumes resources directly. Eliminating the resource cost requires $s_1 = p$, making the technology free and generally inducing over-adoption; any smaller subsidy leaves some resource cost in place. The planner therefore chooses a second-best $s_1 > 0$, trading off the welfare loss from over-adoption at the margin against the resource cost saved on every adoption. The model above could also accommodate a budget-neutral policy if environmental tax revenues are earmarked for an accompanying environmental subsidy.

Correcting an allocation distortion If the principal can induce the same marginal adoptions by correcting an allocation distortion directly, they should do so rather than subsidize the purchase price, as long as the economic cost per induced adoption does not exceed that of the subsidy.

Correcting a resource distortion Suppose the principal can spend resources to reduce r itself. A purchase price subsidy saves $r\tilde{\kappa}$ in financing resources per induced adoption, whereas reducing r saves $(1+r)\tilde{\kappa}$. At equal economic cost per induced adoption, correcting r therefore dominates the purchase price subsidy. Because correction generates greater resource savings, it can remain worthwhile even when its economic cost per induced adoption exceeds that of the subsidy.

Subsidizing a resource distortion Alternatively, the principal can pay $\varsigma(p-s_1)$ of each buyer's resource cost. The buyer then faces the rate $r-\varsigma$ and adopts if and only if

$$v \geq (1+r-\varsigma)(p-s_1).$$

This policy lowers the cost perceived by the buyer but does not eliminate the underlying financing resource cost. Per dollar of expenditure, a purchase price subsidy moves the adoption threshold $1+r$ times as far: each dollar of s_1 reduces both the purchase amount and the financing cost attached to that amount. The purchase price subsidy therefore dominates subsidizing the resource cost on the extensive margin, increasingly so as r rises.

Marginal distortions A marginal distortion of perceived benefits scales the relative fuel price the agent perceives. Adopters underuse the technology, and WTP falls more than proportionally. For the first channel, the distortion still raises the private benefit of the marginal type; however, the distortion also reduces the rate at which private benefits translate into usage. Which of these two effects dominates depends on the nature of the distortion. The second channel, an increase in the demand elasticity, persists under common utility functions as long as there is an increasing hazard rate in types.

Negative correlation between private and social benefits Channel 1 reverses when the conditional external-benefit profile $\Phi(v)$ is decreasing over the relevant range: for example, status buyers who rarely drive adopt an electric vehicle first, while commuters who would displace the most gasoline do not. At the margin the distortion then excludes adopters with the greatest abatement, so a purchase price subsidy generates a lower marginal external benefit than in the undistorted market. The elasticity channel can still favor purchase price subsidies, so the net effect is an empirical question. [Section 7](#) examines it in our context.

Heterogeneous distortions When the distortion is heterogeneous, different instruments no longer induce the same agents. The most distorted agents have the highest adoption thresholds, so an intervention targeted at them induces high-value adopters. A uniform s_1 , on the other hand, is

offered to every agent regardless of their distortion, so it induces the marginal adopter of each distortion type, including low-value adopters among the least distorted. We evaluate this quantitatively in the counterfactual estimation (Section 7).

3 Study setting and experimental design

3.1 Policy background

Residential cooking across the world emits more than 1,200 MtCO_{2e} per year: more than the fleet of U.S. passenger cars and light-duty trucks (1,030 MtCO_{2e}) and the entire U.S. agricultural sector (595 MtCO_{2e}) (Berkouwer and Dean, 2026c). More than a third of cooking-related emissions are generated by the more than two billion people—including 73% of African households—who lack modern cooking technologies and cook primarily with wood and charcoal (IEA, 2024; Berkouwer and Dean, 2026c). At the same time, more than two billion people have gained access to electricity in the past two decades (World Bank, 2024). This has generated widespread policy interest in ‘leapfrogging’ from biomass to electric cooking.

East Africa is at the forefront of this transition: Kenya’s electric utility launched the “Pika na Power” campaign in 2017, and the government’s National Electric Cooking Strategy Action Plan followed with VAT waivers, support for consumer financing, and a dedicated e-cooking tariff (Online Appendix IV.1). Electric cooking is a particularly effective climate mitigation strategy where grids are clean: renewable sources generated 85% of Kenya’s grid electricity in 2023 (Kenya Power, 2023).

On the other hand, voltage fluctuations and power outages might be a concern. In a similar context in Kenya, Wolfram et al. (2027) record an average of 61 minutes of power outages per day and an average voltage of 233 V, 3% below the nominal 240 V. This could dampen the value of electric cooking in this context.

The left panel in Figure 3 shows a standard Kenyan charcoal stove. At baseline, study participants reported using on average 3.1 kilograms of charcoal per day (Appendix Table A1 presents household characteristics). One kilogram of charcoal used in urban Kenya emits approximately 4.4 kilograms of CO_{2e} through production and combustion (Section 3.5 below presents more detail). Each household in our study thus emits approximately 4.9 tons of CO_{2e} per year through its use of charcoal for cooking. For comparison, an average U.S. passenger vehicle emits 4.6 tons of CO₂ per year.¹²

3.2 The ECOA induction stove

The right panel in Figure 3 shows the induction stove we study, manufactured by ECOA in Ruiru, Kenya. The induction stove has a power rating of 200–2,000 W depending on the desired functionality (from low simmer to high boil) and a voltage rating of 240–250 VAC at 50–60 Hz.

¹²The average U.S. passenger vehicle is driven 11,500 miles per year at 22.2 miles per gallon of gasoline, emitting 8.9 kilograms of CO₂ per gallon (EPA, 2025).

Figure 3: Charcoal stove and induction stove



Notes: On the left is a charcoal stove instrumented with a high-frequency temperature sensor. On the right is the induction stove bundle we study.

According to the manufacturer’s internal data, the ECOA induction stove has an expected lifespan of 2.2 years (Online Appendix Figure B1 shows the survival curve). Throughout the rest of the paper, we use this estimate when calculating lifetime abatement and benefits. Technology abandonment in this context could be due to a combination of factors, including not only breakage but also non-payment, which triggers a remote shutdown of the stove (the results discuss how this could mechanically lower stove usage).

At the time of our study launch in August 2025, the cookstove manufacturer had sold more than 50,000 induction stoves in Tanzania and Kenya, primarily through women’s groups and through its direct sales agents. That said, as the stoves were not yet available at retail outlets in our study area, our study population was generally not familiar with the product. The stove with one pot was priced at US\$72 in Kenya. We offered participants a bundle consisting of the stove plus two pots and one frying pan, priced at US\$84. In line with the manufacturer’s social mission, the price equals the marginal production cost: throughout the paper the stove price and the manufacturing cost of the technology are both US\$84.

Residential electricity customers in Kenya face a version of an increasing block tariff. At the start of our study, customers consuming on average less than 30 kWh, 30–100 kWh, or more than 100 kWh per month, respectively, paid US\$0.166, US\$0.20, or US\$0.22 per kWh.¹³ In our sample, 86% consume less than 30 kWh per month and 99% consume less than 100 kWh per month. We use the lowest tariff band in calculations; the resulting estimates align well with estimates that use self-reported electricity spending data.

The manufacturer uses relatively sophisticated credit technologies to help alleviate credit market failures (Online Appendix IV.2 provides more detail). To cover the associated administrative costs, buyers paying in installments under the standard commercial plan must pay back 175% of the principal: an initial down payment of US\$28 is followed by six monthly payments of US\$16 each for

¹³Kenya’s electricity tariff consists of a fixed component plus a floating component pegged to international fuel prices. The total price per kWh for customers in the top band fluctuated between US\$0.2129 and US\$0.2219 over the study period.

a total price of US\$126.¹⁴ Participants can repay their loans early if they wish, but this does not reduce the total payment amount. The payment structure corresponds to an annualized percentage rate (APR) of 224%, well above Kenya’s central bank rate of 9.5% at the time of the main survey (CBK, 2026).

3.3 Data collection

We implemented research activities in peri-urban neighborhoods around the cities of Naivasha, Gilgil, and Nakuru in Nakuru County. Nakuru County is relatively representative of Kenya in terms of its fuel mix: 63% of households cook with firewood or charcoal and 65% have access to electricity, compared with 64% and 58% nationwide, respectively (KNBS, 2024). Participants were visited up to four times (Appendix Figure A1 presents a timeline of study activities). We discuss the four visits first and then provide more information about the randomized treatments.

Baseline survey Study enumerators enrolled 2,503 participants while walking around the study neighborhoods (Appendix Table A1 presents summary statistics). During enrollment, enumerators asked a small number of demographic and socioeconomic questions. To qualify for study participation, participants had to own a standard Kenyan charcoal stove; spend at least half of their cooking fuel budget, and at least US\$3 per week, on charcoal; be at least 18 years of age; have a prepaid Kenya Power electricity meter in their homes; have a household size of at least three individuals; have basic literacy; and not currently own an electric stove or an energy-efficient charcoal stove.

Enumerators also installed temperature sensors on participants’ charcoal stoves, which would continue to collect 2-minute interval data for up to six months after installation. We convert temperature sensor data to a panel dataset of charcoal cookstove usage time by participant. Online Appendix IV.6 provides more detail on the sensors, their installation, and the data processing.

Main survey 2,134 participants completed the main survey, which was conducted around 50 days after the baseline visit (since attrition at this stage occurred before randomized assignment was revealed, it should not affect outcomes). Enumerators provided basic information about the induction stove and implemented the random treatments. They also elicited willingness-to-pay (WTP) in an incentive-compatible manner using a Becker, DeGroot, and Marschak (BDM; 1964) mechanism (Online Appendix IV.3 provides more detail on mechanism implementation).¹⁵

Participants whose WTP was at least their randomly assigned price were expected to purchase the induction stove during the main visit (98% complied). They were then offered a standard 30-

¹⁴US\$98 in installments covers the US\$56 that remains of the US\$84 price after paying the US\$28 down payment.

¹⁵The randomized prices ranged from US\$20 to US\$73, while the BDM elicited WTP between US\$12 and US\$139. Thus, the mechanism was not strictly incentive compatible over small windows at the bottom and top of the WTP distribution. However, participants were not explicitly informed of the random price generation process. They were told by enumerators, “I do not know what the price is: it was randomly chosen using a computer by one of my colleagues. As you can see, the envelope is closed and has your name on it. The price may be lower than the price in the stores! We promise that it will not be higher than what you would pay in a store.” Furthermore, most analyses presented below require only that WTP is incentive compatible on the [20,73] range, which it was.

minute training on how to use the stove and how to contact customer service representatives in their area. Upon installation, the electric stoves began recording energy use in kilowatt-hours (kWh) at 15-minute intervals. This was transmitted wirelessly to remote servers in real time, allowing us to subsidize the electricity used for the induction stove.

Endline surveys 2,081 of the 2,134 main survey participants completed the endline survey, which was conducted on average 40 days after the main survey. There is no evidence of biased attrition that would move our results meaningfully.¹⁶ The endline survey asked primarily about energy expenditures and cooking behavior, as well as other socioeconomic and health outcomes. Enumerators again briefly visited participants around six months after the initial sensor deployment to download temperature data.

Loan repayment For participants who bought the stove with a loan, we observe daily payment data. These track repayment and also allow us to identify the days on which the participant was ‘locked out’ of their stove: this happened if they fell behind on their payments by at least one week.

Market price survey We collect a panel dataset of charcoal prices by conducting market surveys over the course of the experiment across all three cities. This allows us to convert charcoal expenditure data to kilograms of charcoal purchased.

3.4 Randomized treatments

Between the baseline and main surveys, each participant was assigned to several randomized treatments (Appendix Figure A2 presents the randomization tree). All treatments were fully cross-randomized and -stratified. Treatment assignment was furthermore stratified by baseline ownership of LPG, baseline electricity expenditure, and baseline charcoal expenditure, and is balanced on observable baseline characteristics (Online Appendix Table B3).

Purchase price (fixed-cost) subsidy Participants were randomly assigned different subsidy amounts for the fixed cost of buying the induction stove bundle. Relative to the US\$84 price, 49% of participants received a discount of 13% (a price of US\$73) and 46% a discount of 76% (a price of US\$20).¹⁷

Usage (marginal-cost) subsidy Participants in the usage subsidy treatment groups were offered a subsidy on the cost of the electricity that was used to operate the induction stove. One-third of

¹⁶Stove adopters are 2.2 percentage points less likely to attrit, but attrition is otherwise balanced across random treatment assignment (Online Appendix Table B1). 68% of attriters had moved outside the study area (Online Appendix Table B2).

¹⁷The remainder of participants had discounts of 46% (0.2% of participants), 50% (2.5%), or 69% (1.8%). All prices in the feasible range from US\$20 (since a down payment of US\$12 plus at least some repayment was required by the stove firm) to US\$73 (for ethical reasons all study participants received a discount of at least 13%) had positive probability of being selected.

participants were not offered a usage subsidy, one-third were offered a subsidy of 25%, and one-third were offered a subsidy of 75%. Only participants who bought the induction stove would actually receive the subsidy.

During the main visit, prior to the WTP elicitation, recipients were informed of the level and duration of subsidy they would receive and were told that only electricity used by the electric stove was subsidized. For a sense of magnitude, given realized usage averaging 17 kWh per month at a tariff of US\$0.166, total usage subsidy payouts over the six-month subsidy period averaged US\$4.3 and US\$13 for participants in the 25% and 75% subsidy groups, respectively.

Subsidies were paid out weekly via SMS using a unique, non-transferable token ID (Online Appendix IV.4 provides more detail on the implementation methodology). This mirrors standard practice, as participants normally buy tokens using a mobile phone and then enter the token ID into their prepaid meter: 30% of participants normally do this once per week, and 66% buy tokens every 3–15 days. This is also how the electric utility normally implements subsidy programs. During the endline survey, 86% of participants reported that they had been receiving the SMS messages containing the token IDs, and 96% of them reported that they had successfully entered the tokens into their electricity meters.

Payment structure Half of participants were randomly assigned to pay in cash and half to pay with an interest-free loan. Participants paying with cash had to pay the full (randomly assigned) price during the main visit.

Participants paying with credit had to pay a down payment of US\$12.¹⁸ Starting one month after the purchase date, the total amount due increased daily for 90 days. If a participant did not meet the amount due, the stove would shut off seven days later until they made another payment, at which point the amount due would be recalculated from the new payment date (Online Appendix IV.2 provides more detail on the credit policy). Participants could pay early, but this would not reduce the total amount owed.

Those required to pay in cash could have declined our offer and purchased the stove from the stove manufacturer directly using the commercially available payment plan. However, at an APR of 224%, the payment plan would require a total WTP of US\$126; only five (out of 1,040) respondents in the cash group had a WTP at least as high as this.

Information treatment All participants received two information sheets that showed what meals they could cook with a charcoal stove and with an electric stove (Appendix Figure A3). 86% of participants were assigned to an information treatment group: their sheets listed the fuel cost of cooking various meals on an induction stove and on a charcoal stove. For participants in the usage subsidy groups, the information sheets showed costs with and without the subsidy.

¹⁸On the first two days of the main survey, September 23 and 24, a down payment of US\$20 was used due to a miscommunication with the study partner. Excluding surveys collected on these days (5% of the sample) does not meaningfully affect results.

3.5 Converting energy usage to CO₂e emissions

Charcoal We assume a fraction of non-renewable biomass (fNRB)—the fraction of harvested biomass that is not regrown—of 38%, reflecting the most recently modeled fuelwood estimates for Kenya (Ghilardi and Bailis, 2024), though our conclusions are relatively stable even when assuming an fNRB of 10% or 100% (scenarios 3 and 4 of Appendix Table A3). We convert estimates of production, processing, and combustion emissions of CO₂, CH₄, and N₂O to CO₂-equivalent emissions assuming a 50-year time horizon. Online Appendix IV.5 provides more detail.

Electricity Renewable sources of energy generate 85% of Kenya’s annual grid electricity (Kenya Power, 2023), including geothermal (45%), hydroelectric (19%), and wind (17%). To convert the remaining 15% of generation to CO₂e emissions we assume an emissions factor of 1 tCO₂e per MWh of electricity, resulting in an average emissions intensity of 154 g CO₂e per kWh. As a robustness check we show that the results would change only slightly even if we assumed Kenya’s grid was powered 100% by coal. Still, all recent capacity additions in Kenya have been renewable, so we think the average emissions intensity is the relevant long-run benchmark for the clean cooking transition.

Liquefied Petroleum Gas We convert LPG expenditures to cylinders purchased using local prices, and convert cylinders to CO₂e emissions using a conversion ratio of 4.6 kilograms of CO₂e per kilogram of LPG refill.¹⁹

Firewood 14% of participants reported ever using firewood to cook, but we track neither the quantity of wood collected nor firewood prices. Due to the uncertainty in the origin of firewood in this region, we refrain from converting the reduction in firewood usage to CO₂e emissions or estimating any CO₂e abatement generated by firewood reductions. Including this abatement would lower the abatement cost estimates.

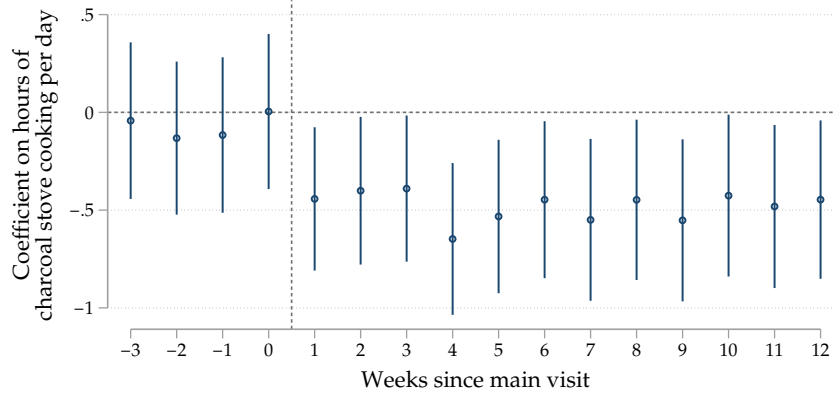
4 Private and social impacts of electric stoves

Of the 2,134 participants who successfully completed the main visit, 626 bought the induction stove, of whom 421 paid with credit. Since actual stove purchasing was not randomly assigned, we use the randomly assigned subsidy, the randomly assigned credit access, and their interaction as instrumental variables to estimate the causal impacts of induction stove adoption on private and social outcomes. The first stage consistently yields an F -statistic above 40.²⁰ Online Appendix V presents all outcomes in the context of the pre-analysis plan.

¹⁹One refill of a 6 kg cylinder of LPG in East Africa emits 27.8 kg CO₂e, or 4.6 kg CO₂e per kg LPG (USAID, 2021). This includes 3.0 kg CO₂e during combustion, 1.2 kg CO₂e during production, and 0.4 kg CO₂e during transport and raw materials production. A typical 6 kg LPG cylinder in Kenya costs approximately Ksh 1,400.

²⁰56% of participants with a high subsidy and access to credit bought the stove, whereas 1% of participants with a low subsidy and paying with cash bought the stove. 23% of participants with a low subsidy and access to credit bought the stove. 39% of participants with a high subsidy and paying with cash bought the stove.

Figure 4: Impact of electric stove purchase on charcoal cooking recorded by temperature sensors (instrumental variables regression)



Notes: Coefficients from an instrumental variables regression. Daily hours of charcoal stove usage derived from temperature sensor data (algorithm described in Online Appendix IV.6). Week markers are defined relative to the main visit: week 0 is the week preceding the visit (days -7 through -1); week 1 begins on the day of the visit (days 0 through 6). Each stove-ownership-by-event-week indicator is instrumented by the randomized price, the credit treatment, and their interaction, each interacted with the corresponding event-week indicator. The regression includes event week and calendar week fixed effects. Standard errors are clustered by participant. Column 1 of Panel A of Table 2 shows the aggregate post period regression estimate using hourly data (Appendix Table A2 shows additional alternative specifications).

4.1 Charcoal stove usage

Figure 4 plots instrumental variables coefficients of the impact of electric stove adoption on average daily hours of charcoal cooking from four weeks before the main visit through twelve weeks after, as recorded by the temperature sensors. Column 1 of Panel A in Table 2 reports the average treatment effect in the post period from a regression using hourly data. The total daily time spent cooking with a charcoal stove decreases by 30 minutes per day, which corresponds to a 33% reduction.

4.2 Energy expenditures

Panel A of Table 2 presents estimates of the causal effect of stove adoption on energy expenditures. Column 2 shows that self-reported monthly charcoal expenditures decrease by US\$9.7 (64%). This is a local average treatment effect estimated among all compliers in the instrumental variables approach. Thus, it factors in behaviors like ‘stacking’ (where participants continue to use their old stoves) as well as participants choosing not to use their induction stoves. Without those behaviors, the estimated charcoal reduction would likely have been higher.

Column 5 shows that self-reported monthly electricity expenditures increase by US\$2.5 (70%).²¹ It is reassuring that this is close to US\$2.9, which is the induction stoves’ average recorded consumption (17 kWh per month) multiplied by the Kenya Power tariff (US\$0.166 per kWh). There

²¹To address the confounding effect of the usage subsidies, column 5 defines total electricity spending as self-reported electricity spending plus subsidy receipts. Column 6 estimates self-reported electricity spending among the subsidy control group.

Table 2: Impact of induction stove purchase (instrumental variables)

A. Energy usage

	Hours per day cooking with charcoal	Monthly expenditures (USD)					
		Charcoal	LPG	Wood	Electricity	Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Bought induction stove (=1)	-0.5*** (0.2)	-9.7*** (0.7)	-0.5 (0.5)	-0.3* (0.2)	2.5*** (0.3)	2.6*** (0.6)	-8.1*** (0.9)
Observations	5370250	2081	2081	2081	2081	703	2081
Control Mean	1.5	15.1	5.5	0.7	3.5	3.7	24.9
F-Stat	135	261	261	261	261	70	261

B. Emissions (tons of CO₂e per year)

	Charcoal	LPG	Electricity	Total
	(1)	(2)	(3)	(4)
	(1)	(2)	(3)	(4)
Bought induction stove (=1)	-2.42*** (0.18)	-0.02 (0.02)	0.03*** (0.00)	-2.41*** (0.18)
Observations	2081	2081	2081	2081
Control Mean	3.77	0.17	0.04	3.99
F-Stat	260.63	260.63	260.63	260.63

Notes: Instrumental variables estimates use the randomized treatment price, the randomized credit condition, and their interaction as exogenous instruments. All regressions control for stratification and treatment indicators, and basic socioeconomic and survey controls. LPG is liquefied petroleum gas. Panel A shows the impact of induction stove purchasing on daily cooking and monthly energy expenditures. Column 1 uses hourly temperature sensor data (standard errors clustered by participant; additional specifications in Appendix Table A2). Columns 2–7 and Panel B report heteroskedasticity-robust standard errors in parentheses. Columns 5 and 7 add subsidies to electricity expenditure. Column 6 estimates self-reported electricity spending among the subsidy control group. Using only price as the exogenous instrument and controlling for willingness-to-pay does not change the results qualitatively (Online Appendix Table B4). Panel B shows the impact on emissions. Online Appendix Table B5 and Online Appendix Table B6 estimate treatment effect heterogeneity with respect to Column 7 of Panel A. Section 3.5 describes the methodology for converting energy expenditures to emissions. ‘Control mean’ evaluated among non-adopters in the complier sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

are additional, smaller decreases in wood and LPG expenditure. Aggregate monthly energy expenditures decrease by US\$8.1 (32%). Assuming a 10% annual discount rate, this corresponds to US\$189 over the stove’s 2.2-year expected usage duration.

At the stove’s price of US\$84, the savings generate a return on investment (ROI) of 101% per year. Even under the commercial installment plan that the manufacturer normally offers, described in Section 3.2, which has an 224% APR, the ROI is 63%. However, the terms of that loan can still be cost-prohibitive. For most customers, the energy savings do not cover the US\$16 monthly payment requirement, and the ROI we estimate—while large—is lower than the payment plan’s interest rate, which reflects the economic costs of administering credit programs in this context. All of this together generates the inefficient under-adoption that we observe. For comparison, the high- and low-subsidy payment plans in the experiment required monthly payments of US\$3 and US\$21,

respectively.

The only baseline socioeconomic characteristic along which we see treatment effect heterogeneity is ownership of an LPG stove: non-owners save US\$11 per month while owners save US\$7 per month (Online Appendix Table B5). The p -value for this comparison is 0.036, and the difference is not statistically significant after adjusting for multiple-hypothesis testing.

4.3 Emissions

Panel B of Table 2 presents the impact on household emissions. Induction stove adoption increases household grid emissions by 0.03 tCO₂e per year. This is a relatively small increase: for comparison, annual emissions from charcoal cooking average 4.9 tCO₂e and from LPG cooking average 0.3 tCO₂e (Appendix Table A1). Charcoal is a highly inefficient source of energy, losing most of the wood’s energy across the kiln and the stove, whereas electricity incurs almost no losses at the induction stove. A much larger share of primary energy therefore becomes cooking heat. While Kenya’s 85% renewable grid helps lower emissions slightly, electric cooking would still only emit 0.16 tCO₂e per year even if the grid were entirely powered by coal (emitting 1 ton of CO₂e per MWh).

Column 4 reports an annual emissions reduction of 2.4 tCO₂e. This effectively corresponds to the reduction in charcoal-related emissions, since the modest increase in electricity-related emissions is offset by a modest, and not statistically significant, decrease in LPG-related emissions (as discussed in Section 3.5, we do not quantify fuelwood reductions).

4.4 Co-benefits and constraints

We briefly discuss several other drivers and impacts of adoption, which are important from a welfare perspective. However, they affect the subsidy cost per tCO₂e only to the extent that they move WTP and usage directly, which our measurements already incorporate. They are furthermore difficult to quantify, and we therefore do not include them in the aggregate welfare calculations. Berkouwer and Dean (2026c) presents additional detail on these economic and institutional factors.

Diet Induction stove adoption generates a 12-percentage-point decrease in the fraction of participants who cook ugali, mukimo, or matoke (three starch staples) and an 11-percentage-point increase in the fraction of participants who cook rice. Anecdotally, rice is easier to cook on an electric stove than other starches. We observe no other meaningful changes in which foods participants cook or in the number of unique foods they cook. We detect no impact on total daily cooking time; the 29-minute-per-day decrease in charcoal cookstove usage is accompanied by the 35 minutes per day of electric cooking, with very little simultaneous use (columns 7–9 of Appendix Table A2).

Health We see meaningful reductions in self-reported cough and breathlessness among participants and their children. In a similar context, Berkouwer and Dean (2026a) find health improvements of similar magnitude. While important for welfare, those improvements did not generate long-term clinical health or economic benefits.

5 Green subsidy efficacy

We begin by evaluating subsidy efficacy among participants without access to an interest-free loan, which best reflects the setting outside our experiment. [Section 6](#) evaluates how subsidy efficacy changes when such a loan relaxes demand distortions.

5.1 Purchase price subsidies

Subsidy cost per marginal adoption We randomly applied a high or low subsidy to the competitive market price $p = \text{US\$84}$, such that $p - s_h = \text{US\$20}$ and $p - s_l = \text{US\$73}$. The willingness-to-pay (WTP) measures elicited using the BDM mechanism indicate that $q_h = 40\%$ of participants would buy the stove with a high subsidy and $q_l = 2.6\%$ would buy the stove with a low subsidy. The principal therefore spends $\mathbb{C}_s = \text{US\$67}$ in additional subsidy expenditures to incentivize one adoption when they increase the subsidy, where:

$$\mathbb{C}_s = \frac{\text{Additional subsidy cost}}{\text{Additional adoptions}} = \frac{q_h s_h - q_l s_l}{q_h - q_l}$$

The causal estimates from the instrumental variables (IV) regression were identified off the compliers, who have a WTP between the low and high prices and are thus exactly marginal to the subsidy increase. We can therefore combine these estimates to evaluate subsidy efficacy.

Benefit per marginal adoption From [Section 4](#), marginal buyers of induction stoves in our context save $\text{US\$9.7}$ per month and abate 2.6 tCO_2e per year (columns 8 and 10 of [Table 4](#)). Over the stove's 2.2-year lifespan, each marginal adopter generates aggregate private benefit of $\hat{v} = \text{US\$228}$ (discounted at 10% per year) and abates 5.7 tCO_2e . Using a social cost of carbon (SCC) of $\text{US\$120}$ (the 2020 Environmental Protection Agency estimate under a 2.5% discount rate; EPA, [2023](#)), this corresponds to a marginal externality benefit of $\hat{\Phi} = \text{US\$688}$.

Subsidy cost per tCO_2e An environmentally minded principal wishes to maximize CO_2e abatement while minimizing subsidy expenditure. This could be a budget-constrained government agency, or a cost-minimizing company or organization with an abatement target. The additional subsidy expenditure required to abate one additional ton of CO_2e is then given by:

$$\text{Subsidy cost per tCO}_2e = \frac{\text{Subsidy cost per additional adoption}}{\text{Tons of CO}_2e \text{ abated per adoption}} = \frac{\text{US\$67}}{5.7 \text{ tCO}_2e} \quad (12)$$

The subsidy cost of abatement in this context is therefore $\text{US\$12}$ per tCO_2e .

Observable welfare gain per subsidy dollar A social principal values all costs or benefits induced by the subsidy. Since WTP is distorted in this context, we cannot use it to measure welfare. We instead evaluate the observable components of welfare per subsidy dollar, which follow from the welfare expression $\mathcal{W}(r)$ in [Section 2.2](#). In this context, this comprises fuel savings $\hat{v}$, plus the value

of the emissions avoided $\hat{\Phi}$, minus the production cost of the electric stove p (conservatively, we leave out the manufacturing cost of the traditional stove), net of any resource costs caused by the distortion, $r(p - \mathbb{C}_s)$:

$$\text{Benefit per subsidy dollar} = \frac{\text{Benefit per marginal adoption}}{\text{Subsidy per marginal adoption}} = \frac{\hat{v} + \hat{\Phi} - p - r(p - \mathbb{C}_s)}{\mathbb{C}_s} \quad (13)$$

This yields an observable welfare gain of between US\$10.2 and US\$12.1 per subsidy dollar, consisting of US\$3.4 in private fuel savings and US\$10 in environmental benefit, minus stove and resource costs. The interval reflects uncertainty about the degree to which the part of the distortion that was not alleviated through the randomized interest-free loan generates resource cost (Section 6 derives the resource cost calculations in more detail).

Alternative measures of subsidy efficacy

Marginal Value of Public Funds (MVPF) We compute the MVPF (Hendren and Sprung-Keyser, 2020) as Eq. 13 plus US\$1 to capture the private benefit of the cash transfer. The MVPF in our context is thus between US\$11.2 and US\$13.1.

Economic cost of abatement Since fuel savings exceed the stove’s manufacturing cost, the economic cost per ton (also known as the resource cost per ton) is negative: in this context, carbon abatement generates economic benefit rather than cost. Each ton of CO_{2e} abated generates between US\$0 and US\$23 in non-environmental economic benefit (fuel savings minus stove manufacturing cost and distortionary resource costs).

Abatement per stove sold Adoption of an additional stove abates 2.6 tCO_{2e} per year, and 93% of sales are additional to a subsidy. One stove sale therefore abates 2.5 tCO_{2e} per year in expectation, or 5.4 tCO_{2e} over the stove’s lifetime. This may be relevant for funders who can only contract on the total number of stoves sold, for example when the principal contracts carbon credit issuances on aggregate sales.

Results under alternative assumptions The benchmark scenario evaluated in this paper uses 2.2 years of stove usage (Online Appendix Figure B1). If the stove’s lifespan were on average one year or three years, the subsidy cost per tCO_{2e} among participants without access to credit would change to US\$26 or US\$8.6, respectively (scenarios 1 and 2 of Appendix Table A3).

The benchmark scenario uses a fraction of non-renewable biomass (fNRB) of 38%. Evaluating subsidy cost at 10% or 100% moves the subsidy cost per tCO_{2e} to US\$14 or US\$9, respectively (scenarios 3 and 4). If Kenya’s grid were fully powered by coal, the cost would increase by less than US\$1: the electric stoves convert energy to heat very efficiently, so their electricity use is generally very low (scenario 5).

Comparison with other technologies These estimates are substantially more favorable than those of many other green technologies. Panel A of Table 3 summarizes the subsidy cost per

Table 3: Efficiency of selected green subsidies

Technology	Observable welfare gain per US\$1 subsidy (US\$)	Subsidy cost per tCO _{2e} (US\$)	Source
<i>A. Low- and middle-income contexts</i>			
Electric stoves	10.2–12.1	12	<i>This paper</i>
Solar lanterns	12	10	Rom et al. (2025)
Energy-efficient motors	4.1	– ^b	Chaurey et al. (2025)
Improved charcoal stoves	26	7	Berkouwer and Dean (2026b)
<i>B. High-income contexts^a</i>			
Hybrid vehicles	0	5,940	Hahn et al. (2026)
Electric vehicles	0.4	1,356	Hahn et al. (2026)
Electric vehicles	0.3	1,202	Allcott et al. (2026)
Vehicle retirement	0	876	Hahn et al. (2026)
Weatherization	0	779	Hahn et al. (2026)
Appliance rebates	0.2	474	Hahn et al. (2026)
Residential solar	2.9	90	Hahn et al. (2026)
Wind production credits	4.9	46	Hahn et al. (2026)

Notes: ^aWe conservatively assume a social cost of carbon (SCC) of US\$120; Hahn et al. (2026) assume US\$193.

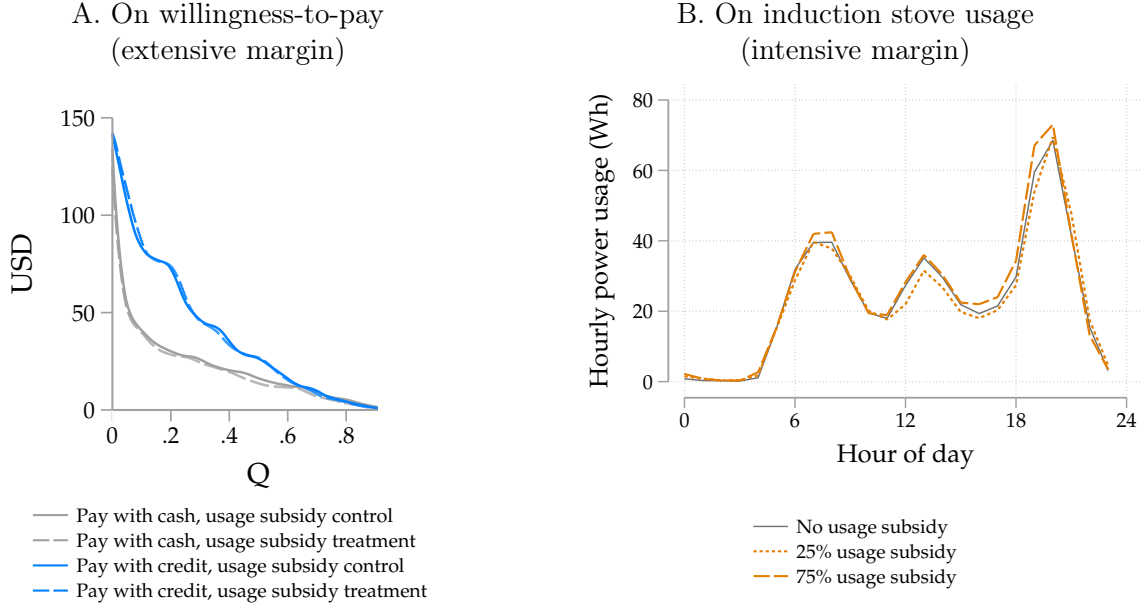
^bChaurey et al. (2025) estimate the impact of disseminating energy-efficient motors for free to non-adopters. We were unable to calculate the subsidy cost per tCO_{2e} from the numbers provided in the manuscript.

tCO_{2e} estimated in this paper as well as in other papers evaluating energy-efficient technologies in contexts where adopters likely face significant demand distortions. Panel B lists estimates of green technologies receiving subsidies in the U.S. Given the evidence thus far, subsidies for green technologies in low- and middle-income contexts appear to generate larger social benefits per subsidy dollar than similar subsidies for green household technologies in high-income contexts.

5.2 Usage subsidies

The absolute efficacy of purchase price subsidies does not inform their efficacy relative to usage subsidies. Usage subsidies could offer two theoretical advantages over a purchase price subsidy of equivalent net present value. First, they could encourage selection into adoption by higher users. Second, they could incentivize higher usage conditional on adoption. Both channels could increase average usage levels conditional on adoption and thereby increase the efficiency of usage subsidies in terms of CO_{2e} abated per dollar of subsidy expenditure relative to an equivalent purchase price subsidy. On the other hand, if agents are myopic or imperfectly attentive to electricity expenditures, they might undervalue usage subsidies that accrue in the future. This could undermine the efficacy of usage subsidies. The relative efficacy of these two subsidy instruments is thus an empirical question.

Figure 5: Impacts of usage subsidies



Notes: Panel A shows the impact of access to credit and usage subsidies on demand, with demand curves constructed using WTP values elicited through the BDM mechanism (point estimates in Appendix Table A4). Panel B shows electric stove usage in watt-hours (Wh) recorded by the induction stoves, separated by whether the participant was in the usage subsidy control group, the 25% usage subsidy group, or the 75% usage subsidy group (point estimates in Appendix Table A5). Online Appendix Figure B2 presents demand curves that adjust for observed repayment rates. Sample is restricted to the 86% of participants assigned to the information treatment; the results are not sensitive to this restriction.

The randomly assigned usage subsidies lowered the cost of electricity used for cooking by 25% or 75%, with each assigned to one-third of the sample. Since we announced the usage subsidy prior to eliciting WTP, the announcement could have affected usage both through a selection channel and through a direct effect. Figure 5 therefore shows the impact of usage subsidies on both the extensive margin (purchasing the stove) and the intensive margin (using the stove).

Panel A plots respondent WTP by whether they received a usage subsidy (and by their credit treatment status). Pooled regressions and separate regressions by subsidy level show no statistically significant impact on WTP (Appendix Table A4). We can rule out that each US\$1 in usage subsidy has more than a US\$0.2 impact (column 3 of Appendix Table A5).

This result matches what the model would predict given the distortions estimated in this context. Comparing WTP with discounted fuel savings, we estimate a demand distortion of 0.9 (as discussed in more detail in Section 6). Each US\$1 of usage subsidy should therefore raise WTP by only US\$0.10. This predicted pass-through lies inside our confidence interval, while undistorted one-for-one pass-through is rejected. Viewed through the model, the unresponsiveness of WTP to usage subsidies could reflect the same distortion between WTP and fuel savings.

Panel B of Figure 5 presents average hourly induction stove usage by usage subsidy. We estimate an elasticity of -0.07, and we can rule out that a 1% increase in price causes more than a 0.3% decrease in usage (column 7 of Appendix Table A5). The point estimate for the 75% subsidy is

larger than the point estimate for the 25% subsidy but amounts to less than 10% of mean usage and is not statistically different from zero.

Together, these results indicate that purchase price subsidies are more effective in this context than usage subsidies.

6 Demand distortions and subsidy efficacy

How does a demand distortion contribute to the subsidy efficacy numbers calculated above? [Section 6.1](#) first quantifies the demand distortion and the resource cost it generates. [Section 6.2](#) then quantifies how the demand distortion affects the externality generated by the marginal adopter and [Section 6.3](#) quantifies how it changes demand elasticity and the subsidy cost per marginal adoption.

Finally, [Section 6.4](#) combines these results to assess aggregate subsidy efficacy. We estimate that relaxing participants' credit constraints through the interest-free loan increases the subsidy cost of abatement per ton of CO₂e to US\$20, whereas it is US\$12 among those in the control group. This distortion thus decreases the subsidy cost of abatement by 41%. It furthermore increases the welfare gains per subsidy dollar by between 35% and 73%, depending on the source of the remaining distortion.

6.1 Quantifying the demand distortion and its resource cost

Participants paying with cash have an average willingness-to-pay (WTP) of US\$20 ([Appendix Table A4](#)). From [Section 4](#), the stove's average fuel savings are $\hat{v} = \text{US\$189}$ over 2.2 years discounted at 10% per year (we use the pooled savings estimate from [Table 2](#) throughout to have a common benchmark across the cash and credit groups). For each credit condition $c \in \{0, 1\}$, we assume the mean demand distortion $\gamma_c \equiv \mathbb{E}[\gamma_i | c]$ is given by $w\bar{t}p_c = (1 - \gamma_c)\hat{v}$. This yields an estimated demand distortion of $\gamma_0 = 0.9$ among those paying with cash. The observed WTP relative to the stream of savings corresponds to an implicit discount factor of 0.02 per year. Put differently, participants make decisions as if they are indifferent between receiving US\$1 today and receiving US\$63 one year from today.

[Figure 5](#) shows that access to credit in this context increases WTP by 75% ([Appendix Table A4](#)). Participants given access to credit have an average WTP of US\$35, suggesting a demand distortion of $\gamma_1 = 0.81$. Their implicit discount factor is 0.09 per year, under which US\$1 today is worth US\$12 in one year. The difference between these values illustrates the degree to which credit constraints distort decision-making in this context.

We are agnostic as to what drives the remaining gap between fuel savings and WTP. Respondents likely experience a range of allocation distortions such as inattention or misinformation, and resource distortions such as transaction costs or additional credit constraints that were not alleviated by the four-month loan. It is also plausible that we are overestimating the private value of adoption, for example because of risk aversion or steeper discounting.²² This would lower the aggregate welfare

²²Without any risk aversion, the discount factor would need to be 0.02 to account for the entire gap. At a 0.9

numbers, but given the random assignment it would not affect relative subsidy efficacy.

To quantify the resource cost, we allow the aggregate demand distortion to reflect the combined effect of two distortions. First, all participants face a background distortion, γ_1 , the nature of which we do not know. Second, those in the credit control group face an additional resource cost r_c , which the interest-free loan alleviates for the treatment group. We can therefore write the aggregate distortion affecting WTP among the control group as $1 - \gamma_0 = \frac{1 - \gamma_1}{1 + r_c}$. Using our estimates of γ_0 and γ_1 , this implies that $r_c = 80\%$. Reassuringly, this is close to the market rate: the commercial installment plan for the induction stove requires paying 75% interest on the amount borrowed.

The aggregate resource cost depends on the degree to which the background distortion is an allocation distortion or a resource distortion. If it is entirely an allocation distortion, then those with access to an interest-free loan experience no resource cost, while the rest incur a resource cost of r_c times the subsidized price. On the other hand, if the background distortion γ_1 reflects a resource distortion, then we can recover its rate r_r using $1 - \gamma_1 = \frac{1}{1 + r_r}$ and apply this to the remaining portion. Solving this with $\hat{\gamma}_1$ yields a total resource rate of 438% for participants with access to the interest-free loan and 871% for participants without. Since we do not know the composition of the background distortion, we report both bounds.

By the end of data tracking, 271 days after stove sale, participants who had bought the stove with credit had paid on average 86% of the price (Appendix Figure A4).²³

6.2 Channel 1: Selection

We first test the impact of demand distortions on the fuel reduction of the marginal adopter. Throughout, we define a marginal adopter as having a WTP between the low price and the high price, so that they would adopt the stove if and only if offered the low price. There are 387 marginal adopters paying with cash and 378 paying with credit, for a total of 765.

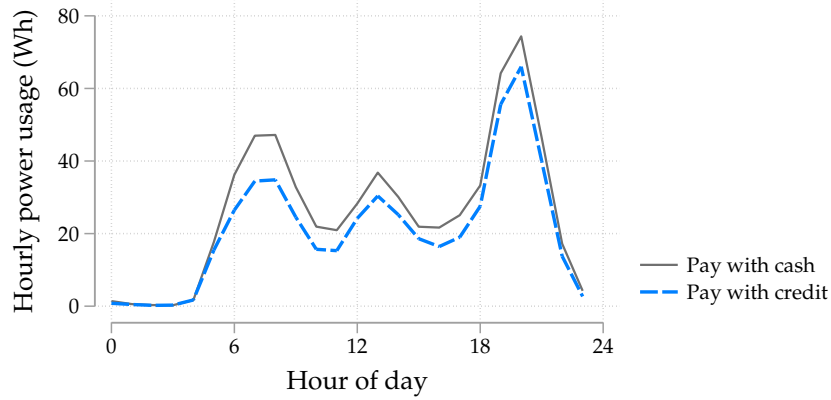
In the context of the model, demand distortions should increase the average usage of agents who are marginal to a subsidy (Section 2.1.1). Figure 6 plots induction stove usage over the course of the day among adopters who were marginal to the subsidy, by credit group. Marginal adopters paying with cash have average usage of 632 Wh per day. This is 24% higher than that of marginal adopters paying with credit, who have average usage of 510 Wh per day.

One might worry that these differences reflect a treatment effect rather than a selection effect: taking on a loan might reduce usage. We explore two plausible explanations in turn. First, credit might lower usage through lock-out: late payment locks the electric stove until the next payment is

discount factor, the net present value of even the lowest decile of estimated TE (US\$95, Online Appendix Table B6) exceeds the WTP of 93% of participants in the credit treatment group; no constant absolute risk aversion (CARA) coefficient can rationalize WTP for them. Factoring in the standard errors of the heterogeneous TE yields a CARA coefficient of 0.052, which is implausibly large: participants would reject a 50–50 gamble between winning US\$100 and losing US\$13. Even at 50% annual discounting, the lowest TE estimate exceeds the WTP of 76% of credit treatment participants. While risk aversion is likely present, it is unlikely to explain most of the remaining gap.

²³Adjusting WTP following observed repayment rates changes the difference in WTP between those with and without access to a loan, but the increase in WTP due to the loan is always large and statistically significant (Online Appendix Table B7).

Figure 6: Electric stove usage among marginal adopters by credit condition



Notes: Hourly usage in Wh by randomized credit condition among marginal adopters, defined as having $p - s_h \leq wtp_i < p - s_l$. Appendix Table A6 presents point estimates (p -val = 0.053).

made, which could lower usage by participants who bought the stove with credit. We cannot test for this by restricting the sample to adopters with low lock-out, since we do not observe counterfactual repayment rates among those who paid with cash and non-payment is highly selected. However, we can limit the sample to adopters who paid the lowest price. In this sample, lock-out rates are low (Online Appendix Figure B3). The result persists with similar magnitude and precision (columns 5 and 6 of Appendix Table A6). Alternatively, one might worry that having to pay back a loan leaves less money available for electricity expenditures. However, the magnitude and precision are also stable when only evaluating the first month after adoption, when adopters paying with cash would have just faced a large direct liquidity shock from adoption (columns 7 and 8 of Appendix Table A6). The difference in usage across adopters with and without credit is therefore likely the result of selection and not a treatment effect.

Table 4 tests whether the difference in electric stove usage translates into differences in aggregate fuel usage. Columns 7 and 8 show that marginal adopters with more distorted demand save on average US\$9.7 per month, whereas adopters whose demand distortion was reduced through access to credit save US\$7.4 per month (p -val = 0.058). These values correspond to US\$228 and US\$175 in total discounted private savings over 2.2 years, respectively.

Because social and private benefits are positively correlated in this context, savings should translate into a similar difference in positive externalities. Columns 9 and 10 of Table 4 show that this is the case. Marginal adopters paying with cash abate 5.7 tCO₂e over 2.2 years, whereas marginal adopters paying with credit abate 4.8 tCO₂e (p -val = 0.048). In other words, the demand distortion in this context increases the abatement caused by marginal adopters by 19%.

6.3 Channel 2: Demand elasticity

The second channel through which demand distortions increase subsidy efficacy is demand elasticity, which can change the number of agents who are marginal to the subsidy and the number of

Table 4: Monthly savings and annual abatement among marginal electric stove adopters

	Charcoal				Electricity		Total			
	(1) USD	(2) USD	(3) CO _{2e}	(4) CO _{2e}	(5) USD	(6) USD	(7) USD	(8) USD	(9) CO _{2e}	(10) CO _{2e}
Credit (=1)	-1.1 (1.4)	-1.1 (1.4)	-0.3 (0.4)	-0.3 (0.4)	0.1 (0.7)	0.1 (0.7)	-0.7 (2.0)	-0.7 (2.0)	-0.2 (0.4)	-0.2 (0.4)
Bought stove (=1)	-10.4*** (0.6)		-2.6*** (0.2)		2.3*** (0.3)		-9.7*** (0.9)		-2.6*** (0.2)	
Bought X No credit		-10.4*** (0.6)		-2.6*** (0.2)		2.3*** (0.3)		-9.7*** (0.9)		-2.6*** (0.2)
Bought X Credit	1.6* (0.8)	-8.9*** (0.6)	0.4* (0.2)	-2.2*** (0.1)	-0.2 (0.4)	2.2*** (0.3)	2.2* (1.2)	-7.4*** (0.8)	0.4** (0.2)	-2.2*** (0.1)
Observations	754	754	754	754	754	754	754	754	754	754
Control Mean	15.7	15.7	3.9	3.9	3.6	3.6	25.5	25.5	4.1	4.1

Notes: Instrumental variables regressions using the randomly assigned price as an instrument for adoption to estimate its impact on spending and emissions among marginal adopters (i.e., $WTP \in [US\$20, US\$73]$). USD amounts are monthly; CO_{2e} amounts are annual. Heteroskedasticity-robust standard errors in parentheses. IV compliers correspond to marginal adopters. Odd columns test whether the marginal adopters' benefits differ by credit condition. Even columns estimate the impact by credit condition. Endline surveys were conducted with 754 of the 765 marginal adopters (98.6%). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

inframarginal payments. This changes the marginal subsidy expenditure to induce one additional adoption. For an experiment with two price points, the cost increase resulting from a subsidy increase can be decomposed as:

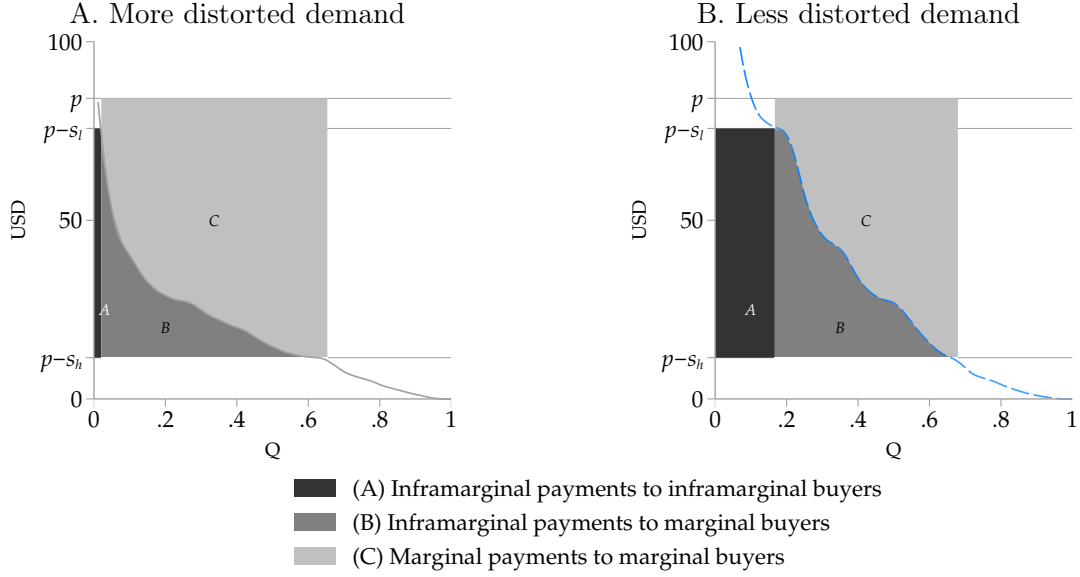
$$q_h s_h - q_l s_l = \underbrace{q_l (s_h - s_l)}_{\text{Increased payouts to inframarginal buyers}} + \underbrace{s_h (q_h - q_l)}_{\text{Full payouts to marginal buyers}} \quad (14)$$

Figure 7 presents intuition on how these effects operate in the context of the demand curves elicited through the BDM. Area *A* represents the first term in Eq. 14 and $B \cup C$ represents the second term. Comparing Panels A and B, demand distortions in this context decrease payouts to inframarginal buyers both in absolute terms and as a fraction of total payouts. 2.6% of participants paying with cash would buy the stove at the high price whereas 21% of participants paying with credit would. Among those paying with credit, 38% of subsidy expenditures flow to inframarginal buyers; among those paying with cash, only 6.5% of expenditures do.

Taken together, the environmental principal incurs a total subsidy cost of US\$96 per additional sale among participants with access to credit and US\$67 among participants without. In other words, relaxing demand distortions increases the subsidy cost of achieving one additional sale by 42%.

These differences reflect a change in the demand elasticity. Table 5 reports the price-quantity coefficients. Column 5 reports a demand elasticity of -2.3 among participants paying with cash, while column 6 reports an elasticity of -0.9 among those paying with credit.

Figure 7: Marginal and inframarginal expenditures by credit condition



Notes: Demand curves constructed using WTP values elicited through the BDM mechanism. Area A is the additional payout to inframarginal buyers. Areas B and C together are the payouts to marginal buyers. Participants having to pay with cash have more distorted demand than participants randomly given access to credit. The subsidy increase induces a larger share of marginal buyers under the more distorted demand curve.

Table 5: Extensive margin demand elasticities by credit treatment group

	Quantity (%)			Log(Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)
Price (USD)	-0.84*** (0.03)	-0.96*** (0.06)	-0.80*** (0.04)			
Log(Price)				-1.23*** (0.05)	-2.33*** (0.13)	-0.88*** (0.05)
Observations	765	387	378	765	387	378
Sample		Pay with cash	Pay with credit		Pay with cash	Pay with credit

Notes: We calculate quantities at different prices over the entire sample (including inframarginal adopters, or always-takers) and then estimate elasticities over the policy-relevant sample of compliers, i.e., anyone with willingness-to-pay between $p-s_h$ and $p-s_l$. Standard errors in parentheses are from 500 bootstrap replications that resample participants within credit treatment group and recompute the demand curve in each replication. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Aggregate impact of demand distortions on subsidy efficacy

Table 6 presents estimates of aggregate subsidy efficacy for participants whose demand was more distorted and less distorted. Its rows map onto key model parameters and empirical estimates. The Channel 1 rows are the empirical counterparts of the average usage of induced adopters, valued at the per-unit fuel saving π and at the per-unit externality ϕ . The Channel 2 rows are the empirical counterparts of demand elasticity and the number of subsidy recipients per induced adoption (ζ). The aggregate subsidy cost per ton of CO₂e abated is US\$20 among participants with access to

Table 6: Impact of demand distortions on purchase subsidy efficacy through the two channels

	Less distorted ($\hat{\gamma}_1 = 0.81$) (1)	More distorted ($\hat{\gamma}_0 = 0.9$) (2)	Advantage of subsidies in a distorted market (3)
A. Empirical estimates			
<i>Channel 1: Marginal adopter</i>			
Fuel savings per marginal adoption ($\hat{v}$; Table 4)	US\$175	US\$228	+30%
Abatement per marginal adoption ($\hat{\Phi}/\text{SCC}$; Table 4)	4.8 tCO ₂ e	5.7 tCO ₂ e	+19%
Externality benefit per marginal adoption ($\hat{\Phi}$)	US\$579	US\$688	+19%
<i>Channel 2: Subsidy cost</i>			
Extensive margin elasticity ($\hat{\mathcal{E}}$; Table 5)	−0.9	−2.3	
Subsidy expenditure to inframarginal adopters	38%	6.5%	
Subsidy expenditure per marginal adoption (C_s)	US\$96	US\$67	−29%
Rate r if γ_1 is an allocation distortion	0	80%	
Rate r if γ_1 is a resource distortion ($= \hat{\gamma}/(1 - \hat{\gamma})$)	438%	871%	
B. Aggregate subsidy efficacy			
Subsidy cost per tCO ₂ e (Eq. 12)	US\$20	US\$12	−41%
<i>Observable welfare gain per subsidy dollar (Eq. 13)</i>			
Remaining distortion γ_1 is 100% allocation	US\$7.0	US\$12.1	+73%
Remaining distortion γ_1 is 100% resource	US\$7.5	US\$10.2	+35%

Notes: ‘Less distorted’ are participants with access to an interest-free loan. ‘More distorted’ are participants who must pay the full price in cash. All calculations use a stove bundle price equal to the production cost ($p = \text{US\$84}$), a social cost of carbon of US\$120 per tCO₂e, and an expected stove life of 2.2 years, and discount fuel savings at 10% per year. Online Appendix Table B8 shows Column 3 with bootstrapped 95% confidence intervals.

credit, whereas it is US\$12 among those with a larger demand distortion. The distortion thus decreases the subsidy cost of abatement by 41%. Bootstrapped standard errors generate a 95% confidence interval for the difference of [28%, 51%] (Online Appendix Table B8).

The observable welfare gain is between US\$7.0 and US\$7.5 per subsidy dollar among participants with access to credit, compared with US\$10.2–US\$12.1 among those with more distorted demand. Credit constraints thus increase the welfare gains per subsidy dollar by between 35% (95% CI: [5.5%, 69%]) and 73% (95% CI: [37%, 115%]), depending on the source of the remaining distortion.

The observable welfare gain expression contains the term $r(C_s - p)$: a resource distortion generates a resource cost rp for induced adopters, but this is offset by the reduction in resource cost rC_s across all adopters. A resource distortion thus causes subsidies to generate a larger welfare gain than an allocation distortion if and only if C_s exceeds p , which equals US\$84 in this context. This holds among participants with credit access, for whom $C_s = \text{US\$96}$, but not for participants paying with cash, for whom $C_s = \text{US\$67}$. The observable welfare gain for the less distorted group

is therefore larger under a resource distortion, but for the more distorted group it is larger under an allocation distortion.

7 Model estimation and counterfactuals

Even among participants who are given access to credit, willingness-to-pay (WTP) of US\$35 is well below the total savings of US\$189 over 2.2 years (savings are discounted at 10% per year). As discussed above, it is possible that our estimates overestimate the private benefit from adoption (Section 6.1). Still, the full stream of financial benefits discounted at 10% per year is a useful benchmark against which to compare the results, and it is reasonable to think that our short intervention did not alleviate all demand distortions experienced by households in this context. It may not have fully alleviated credit constraints: given that average monthly savings were US\$8.1 and the loan had a four-month repayment period, someone with a binding savings constraint might be able to pay only US\$32 over the period of the loan.

To quantify outcomes in alternative scenarios in which the demand distortion is alleviated further, we therefore estimate a generalization of the model specified in Section 2. The counterfactual analyses evaluate how the mean and standard deviation of the demand distortion, and the correlation between private and social benefits, affect subsidy efficacy.

7.1 Setup

We assume agents i are distributed Weibull by their private benefit from adopting: $(v_i/\hat{v}) \sim \text{Weibull}(\Omega_1, \Omega_2)$, where $\hat{v} = \text{US\$189}$ sets the scale of benefits. The Weibull's shape parameter governs tail thickness, allowing the data to discipline how concentrated WTP is around its mean.²⁴

We allow social benefit to be a noisy function of private benefit with a mixing parameter $\Omega_3 \in [0, 1]$ determining the correlation: $\Phi_i = \hat{\Phi} [\Omega_3 (v_i/\hat{v}) + (1 - \Omega_3) \varepsilon_i]$, with ε_i an independent draw from the same Weibull distribution. In a slight shift of notation, in this section we let Φ denote abatement in tons of CO_2e rather than its dollar value, so $\hat{\Phi} = 5.3 \text{ tCO}_2e$.

For each credit condition $c \in \{0, 1\}$ the mean demand distortion is defined by $w\bar{t}p_c = (1 - \gamma_c)\hat{v}$. We assume a heterogeneous demand distortion $\gamma_i = \text{logistic}(\mu_c + \sigma_c \eta_i)$, $\eta_i \sim N(0, 1)$, with dispersion $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \gamma_c)))$, such that Ω_4 governs baseline dispersion and Ω_5 controls how dispersion varies with the mean distortion, allowing heterogeneity in credit impacts to affect the dispersion of WTP.²⁵ The location parameter μ_c is pinned down by $\mathbb{E}[\gamma_i|c] = \gamma_c$. This then yields individual WTP $wtp_i = (1 - \gamma_i) v_i$.

The distributions of wtp_i and Φ_i can be used to calculate demand elasticity $\mathcal{E}_D$ and the average externality of adopters who are marginal to the subsidy ($\bar{\Phi} = \mathbb{E}[\Phi_i | p - s_h \leq wtp_i < p - s_l]$, where $p - s_h = \text{US\$20}$ and $p - s_l = \text{US\$73}$ as in the experiment).

²⁴It satisfies the increasing-hazard-rate condition if $\Omega_1 \geq 1$; we estimate $\hat{\Omega}_1 = 1.9$.

²⁵The logit-normal distribution across different $\mathbb{E}[\gamma]$ reasonably approximates the empirical distributions of γ_i shown in Online Appendix Figure B4. Although the Beta distribution also has support on $[0, 1]$, we do not use it because its behavior at the boundaries of $[0, 1]$ is inconsistent with what we observe in our data.

7.2 Model estimation

We target seven empirical moments—a common correlation between private and social benefits $\nu = \text{corr}(v_i, \Phi_i)$, as well as three moments that vary by credit group: extensive margin elasticity $\mathcal{E}_{D,c}$, average marginal externality $\bar{\Phi}_c$, and cost per marginal adoption. For each moment n we define the simulated moment error $e_n(\mathbf{\Omega})$ as a normalized function of the empirical moment $m_n(x)$ and the simulated moment $\hat{m}_n(\tilde{x})$:

$$e_n(\mathbf{\Omega}) = \frac{\hat{m}_n(\tilde{x}) - m_n(x)}{m_n(x)}.$$

We estimate the model parameters $\mathbf{\Omega} = \{\Omega_1, \Omega_2, \Omega_3, \Omega_4, \Omega_5\}$ using two-step generalized method of moments (GMM) with simulated moments, defining the estimator:

$$\hat{\mathbf{\Omega}}_{GMM} = \arg \min_{\mathbf{\Omega}} e(\mathbf{\Omega})^T \mathbf{W} e(\mathbf{\Omega})$$

We simulate 100,000 agents per parameter evaluation. The first-step weighting matrix is $\mathbf{W}_1 = \mathbf{I}$; the second-step matrix is $\mathbf{W}_2 = \hat{\mathbf{\Gamma}}^{-1}$, where $\hat{\mathbf{\Gamma}}$ is the sample covariance of the simulated moment-error vector across 30 re-simulations at the first-step estimate, holding the empirical moments fixed. The reported estimate corresponds to the lowest objective value across multiple starting values and satisfies the optimizer’s convergence criterion.

The estimated coefficients are $\hat{\Omega}_1 = 1.9$, $\hat{\Omega}_2 = 1$, $\hat{\Omega}_3 = 0.6$, $\hat{\Omega}_4 = -2.3$, and $\hat{\Omega}_5 = 24$. Panel A of [Table 7](#) compares the moments simulated using the estimated parameters with the empirical moments calculated using the data. The subsidy costs per tCO₂e implied by the targeted moments, US\$12 and US\$20, closely match the empirically estimated values of US\$12 and US\$20, respectively. With these estimates in hand, we run counterfactuals by drawing distributions of private benefits $(v_i/\hat{v}) \sim \text{Weibull}(\hat{\Omega}_1, \hat{\Omega}_2)$, social benefit shocks $\varepsilon_i \sim \text{Weibull}(\hat{\Omega}_1, \hat{\Omega}_2)$, and distortions γ_i from the logit-normal distribution with dispersion $\sigma = \log(1 + \exp(\hat{\Omega}_4 + \hat{\Omega}_5(1 - \gamma)))$ and location pinned down by $\mathbb{E}[\gamma_i] = \gamma$.

Counterfactual I: Demand distortion Panel B of [Table 7](#) shows the results implied by the model for different values of γ . Demand distortions that dampen demand increase both the positive externality generated by the marginal adopter and the demand elasticity, lowering the subsidy cost per tCO₂e. In the undistorted case where private benefits equal WTP, the subsidy cost per tCO₂e increases to US\$137.

The model is overidentified with 2 degrees of freedom (the Hansen J statistic is 22.8; $p\text{-val} < 0.001$); however, the counterfactual estimates are not sensitive to model specification: abatement in the undistorted counterfactual is consistently at least ten times as expensive as in the observed case across alternative specifications.²⁶

²⁶For example, Online Appendix Table B11 keeps the logit-normal distortion of the main specification but draws private benefits $v_i/\hat{v}$ and ε_i from a Gamma distribution; Online Appendix Table B12 uses the same distribution of benefits as the main specification but estimates a logit- t distribution instead of a logit-normal distribution. Both

Table 7: Simulated estimates from observed and counterfactual parameters

A. Estimation

Moment	RCT: Pay in cash ($\gamma = 0.9$)			RCT: Pay with credit ($\gamma = 0.81$)		
	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.8	0.8	-1%	0.8	0.8	-1%
Marginal externality ($\bar{\Phi}$)	5.7	5.8	1%	4.8	4.9	2%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.2	6%	-0.9	-0.8	13%
USD per marginal adoption	67.5	67.4	0%	95.5	96.8	1%

B. Counterfactual simulations: Varying the distortion γ

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.9	0.81	0.75	0.5	0.25	0
Marginal externality ($\bar{\Phi}$)	7.9	5.8	4.9	4.6	3.8	3.3	2.9
Extensive margin elasticity ($\mathcal{E}_D$)	-8.2	-2.2	-0.8	-0.5	-0.2	-0.1	-0.1
USD per marginal adoption	64	67	97	125	235	321	397
<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		12	20				
Simulated	8	12	20	27	61	96	137

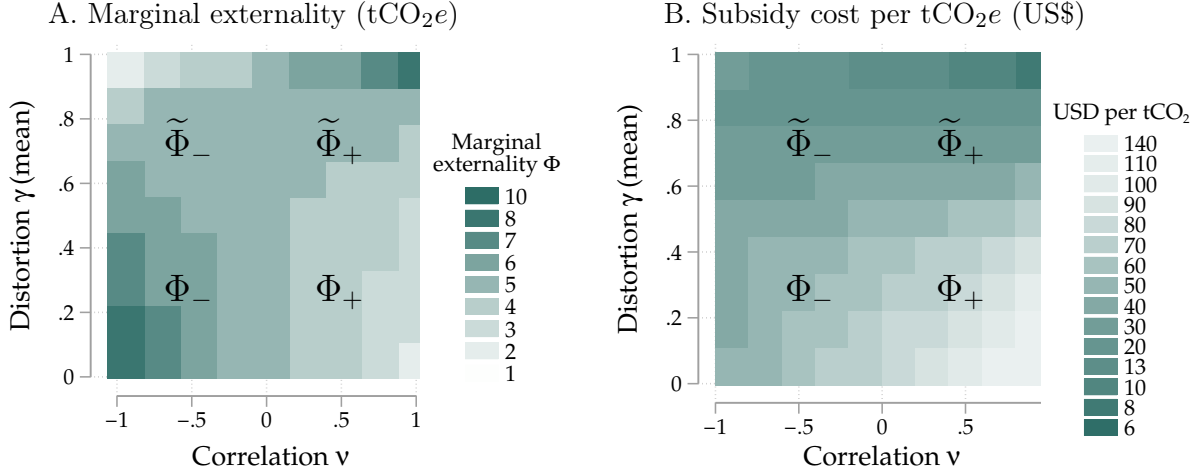
Notes: Panel A compares empirical moments and the simulated moments under the estimated model parameters. There are seven moments: a single correlation moment ν , and one per credit group for each of the other three moments. Panel B uses the model parameters to estimate key moments, as well as the aggregate subsidy cost per tCO₂e, for counterfactual scenarios with larger or smaller demand distortions, holding the correlation between private benefits and social benefits constant at the empirical moment and assuming $r = 0$. Online Appendix Table B9 reports the same estimates with standard errors and confidence intervals. Online Appendix Table B10 also varies the correlation parameter ν . Online Appendix Table B11 and Online Appendix Table B12 show that these results are not sensitive to the distributional assumptions for v_i and γ_i .

Similar patterns persist when estimating the observable welfare gain, Marginal Value of Public Funds (MVPF), and economic cost per ton (Online Appendix Table B13).

Counterfactual II: The correlation between private and social benefits This model also allows us to evaluate how the correlation between private and social benefits affects subsidy efficacy. We are agnostic about what drives this correlation. For example, in another context a clean technology might generate a reduction in pollution exposure that increases WTP even in the absence of any financial savings: this would still cause a positive correlation between private and social benefits, and the model’s insights would still apply.

Figure 8 shows the results when varying the demand distortion γ as well as the correlation ν . Positive values of ν are generated by varying the mixing weight Ω_3 ; negative values are generated by mixing in a draw from the same Weibull distribution that is rank-anti-correlated with $v_i/\hat{v}$. The horizontal axis reports the realized correlation. To help build intuition, we use labels $\tilde{\Phi}_+$, Φ_+ , $\tilde{\Phi}_-$, simulations show similar patterns across all measures of subsidy efficacy.

Figure 8: Counterfactual estimates



Notes: Both panels show how demand distortions as well as the correlation between private and social benefits affect subsidy efficacy. Panel A shows the marginal externality. The labels $\tilde{\Phi}_+$, Φ_+ , $\tilde{\Phi}_-$, and Φ_- correspond to the labels in Panels A and B of Figure 1, which presents the theoretical underpinnings for the selection channel. Panel B shows the combined effect of the marginal externality and the elasticity on the subsidy cost per tCO₂e. The demand distortion decreases the subsidy cost per tCO₂e everywhere in this context, regardless of correlation. Online Appendix Table B10 presents a subset of these estimates in the format of Table 7.

and Φ_- to mark the approximate externalities (also labeled in Panels A and B of Figure 1).

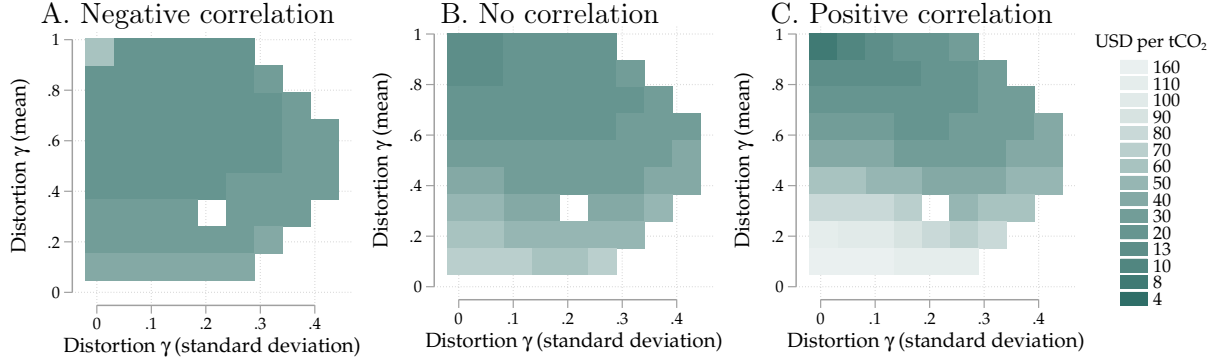
When WTP reflects private benefits and there is no distortion ($\gamma = 0$, such that $wtp_i = v_i$), a positive correlation induces adverse selection, decreasing the marginal positive externality ($\Phi_+ < \Phi_-$). On the other hand, under a demand distortion that dampens WTP ($\gamma > 0$), a positive correlation means compliers generate larger positive externalities, inducing beneficial selection ($\tilde{\Phi}_+ > \tilde{\Phi}_-$). The effect of a demand distortion thus hinges on the correlation between private and social benefits: when this correlation is negative, a demand distortion decreases the average externality of compliers ($\tilde{\Phi}_- < \Phi_-$). Only when the correlation is positive does the distortion increase the average externality of compliers ($\tilde{\Phi}_+ > \Phi_+$).

Panel B shows how both channels together affect the subsidy cost per tCO₂e. While the marginal externality channel depends on whether the correlation is positive or negative, the demand distortion increases the extensive margin elasticity regardless of the correlation between private and social benefits. In this context, the latter effect outweighs the opposing effect on the average externality of compliers for negative correlation coefficients: even though $\tilde{\Phi}_- < \Phi_-$ (Panel A), the distortion still generates a lower subsidy cost per tCO₂e (Panel B). As a result, the demand distortion unambiguously decreases the subsidy cost per tCO₂e, regardless of the correlation coefficient.

Counterfactual III: Heterogeneous demand distortions across agents Finally, we explore how heterogeneity in the demand distortion affects subsidy efficacy. For simplicity we assume the γ_i are uncorrelated with private and social benefits. We draw $\gamma_i = \text{logistic}(\mu + \sigma\eta_i)$, $\eta_i \sim N(0, 1)$, choosing μ and σ to trace out different means and standard deviations of γ_i while ensuring $\gamma_i \in [0, 1]$.

Figure 9 shows the results. When the standard deviation approaches 0, the graphs follow the

Figure 9: Counterfactual estimates with heterogeneous demand distortions



Notes: This figure shows how different demand distortions and correlations affect the subsidy cost per tCO₂e when demand distortions are allowed to be heterogeneous across agents, independently of private and social benefits. Since we constrain $\gamma \in [0, 1]$, not all combinations of mean and standard deviation are feasible. The realized correlations in Panels A and C are -0.94 and 0.94 , the most negative and most positive values attainable given the Weibull marginal distributions.

patterns in Panel B of Figure 8: the mean distortion matters more when private and social benefits are positively correlated. All three panels show that this pattern weakens as the standard deviation increases.

8 Conclusion

Demand distortions such as credit market frictions and behavioral biases are pervasive across many low- and middle-income countries, often dampening willingness-to-pay (WTP) for profitable technologies. This paper first provides theoretical underpinnings for how these types of distortions can increase the observable welfare gain per subsidy dollar and lower the subsidy cost of abatement. This operates through two channels: increasing the private and social benefits of the marginal adopter, and increasing demand elasticity. We characterize conditions under which these gains outweigh the additional resource cost caused by the distortion.

We then test this theory in a field experiment on an energy-efficient technology, induction stoves, with more than 2,000 households in Nakuru County, Kenya, where credit market failures distort demand. We find that the stove abates 5.3 tCO₂e and reduces (discounted) household energy expenditures by US\$189 over its 2.2-year expected lifespan. Even with continued use of charcoal stoves and imperfect additionality of the subsidies, green subsidies in this context abate CO₂e at only US\$12 per ton. This is far below the cost of many other abatement technologies available today and below almost all estimates of the social cost of carbon. Factoring in the private benefit to adopters, the externality benefit, the stove manufacturing cost, and the resource cost, we estimate that each dollar of subsidy generates between US\$10 and US\$12 in observable welfare gain.

Demand distortions in this context partly drive these results. Alleviating the demand distortion lowers the observable welfare gain per subsidy dollar to US\$7–US\$8 and raises the subsidy cost per tCO₂e to US\$20. These empirical estimates align with the two channels identified in the

model: demand distortions increase the marginal adopter’s positive externality by 19% and increase demand elasticity from -0.9 to -2.3 . To simulate subsidy cost per tCO_{2e} in the absence of any demand distortions, we estimate the model using the random variation from the experiment for identification. Counterfactual scenarios indicate that if WTP equaled the discounted stream of fuel savings, the subsidy cost per tCO_{2e} would increase more than tenfold across a range of assumptions and specifications, to US\$137 in our preferred specification.

At annual abatement of 2.6 tCO_{2e}, a widespread cooking transition by all 31 million African households that use charcoal as their primary cooking fuel could abate more than 80 million tons of CO_{2e} per year (Berkouwer and Dean, 2026c). More than 300 million households across the world still cook primarily with firewood, and the IEA (2025a) estimates that the number of households cooking with wood or charcoal in Africa in 2050 will, if anything, be higher than today. Approximately 1.4 million Kenyan households use charcoal on a daily basis, primarily in urban and peri-urban areas, where electricity access was 91% in 2024 (KNBS, 2024). Kenya’s current peak demand is 2,177 MW against installed capacity of 3,243 MW (Kenya Power, 2024). Given our estimates, one million cookstove sales would add only 88 MW to Kenya’s peak load. Other African countries have seen similar growth in electricity access rates and renewable energy generation.

What do these results mean for global climate mitigation more broadly? More than US\$1 trillion is invested in climate change mitigation each year (Buchner, 2024). Maximizing the impact of these expenditures requires allocating this funding towards abatement opportunities at the lowest-cost portion of the abatement cost curve. The U.S. has relatively sophisticated financial markets for consumer durables. Agents with large private benefits already have access to financing to adopt more energy-efficient technologies. This leaves agents with lower average usage, who generate smaller environmental benefits, disproportionately marginal to a subsidy. It also increases inframarginal expenditure as a fraction of total expenditure. Given this paper’s results, many of the world’s most cost-effective abatement opportunities may be found in low- and middle-income countries, where demand distortions are pervasive. These contexts are furthermore expected to drive nearly all of the growth in global energy demand in coming decades (IEA, 2025b). Additional research can help identify specific mitigation opportunities and characterize how the heterogeneity of demand distortion types across countries and sectors affects the optimal deployment of green financing.

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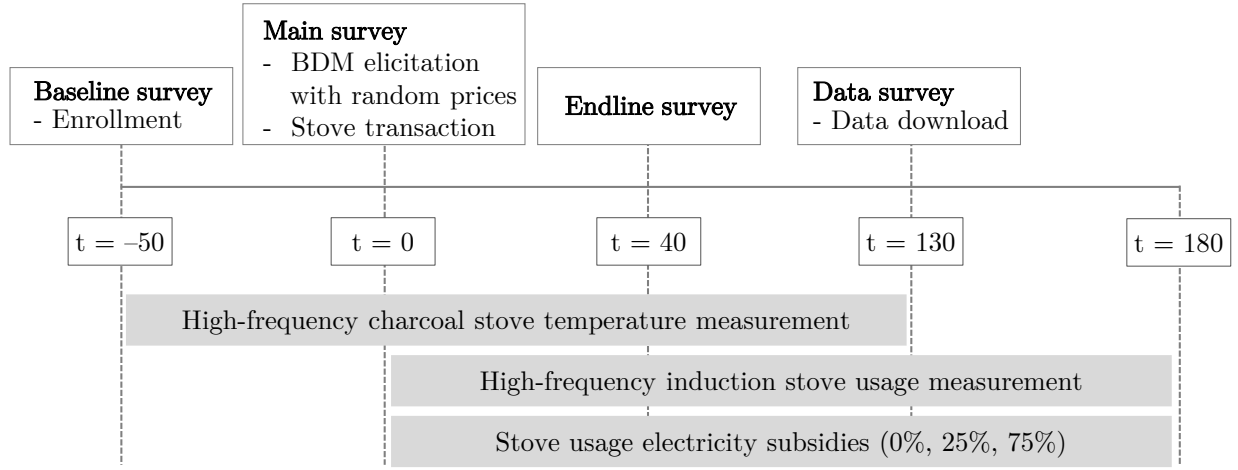
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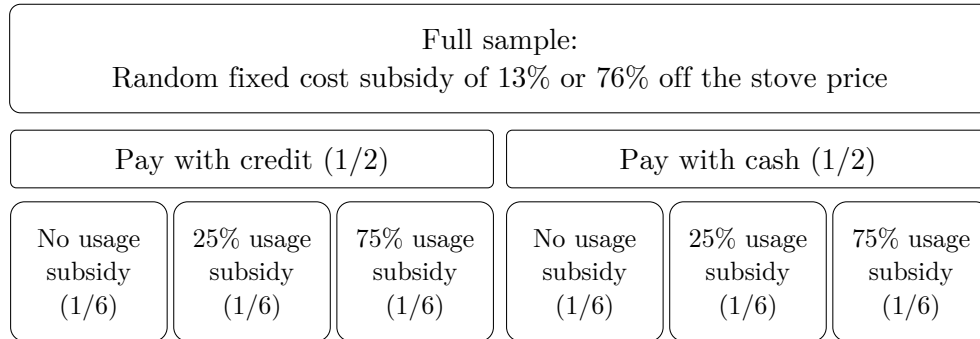
I Appendix figures

Figure A1: Experiment timeline



Notes: Timeline of study activities. The baseline survey was conducted on average 50 days before the main survey ($[10^{th}, 90^{th}]$: $[40, 60]$ days), between August 12 and September 19, 2025. The main survey was conducted between September 23 and November 19, 2025. The endline survey was conducted on average 40 days after the main survey ($[25, 55]$), between November 10 and December 12, 2025. The data download was conducted on average 130 days after the main survey ($[110, 150]$), between February 18 and March 6, 2026. High-frequency induction stove usage monitoring and usage subsidy transfers (as electricity tokens) were completed remotely.

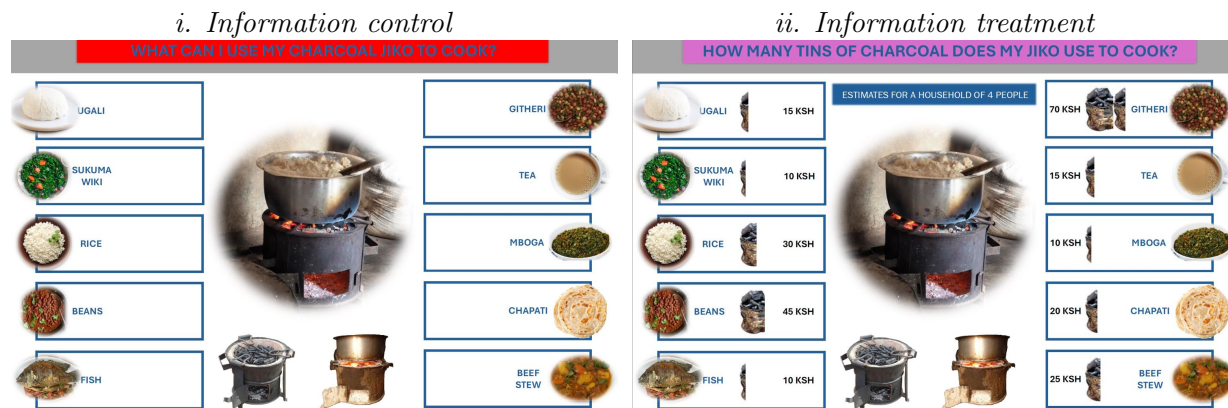
Figure A2: Randomized treatments



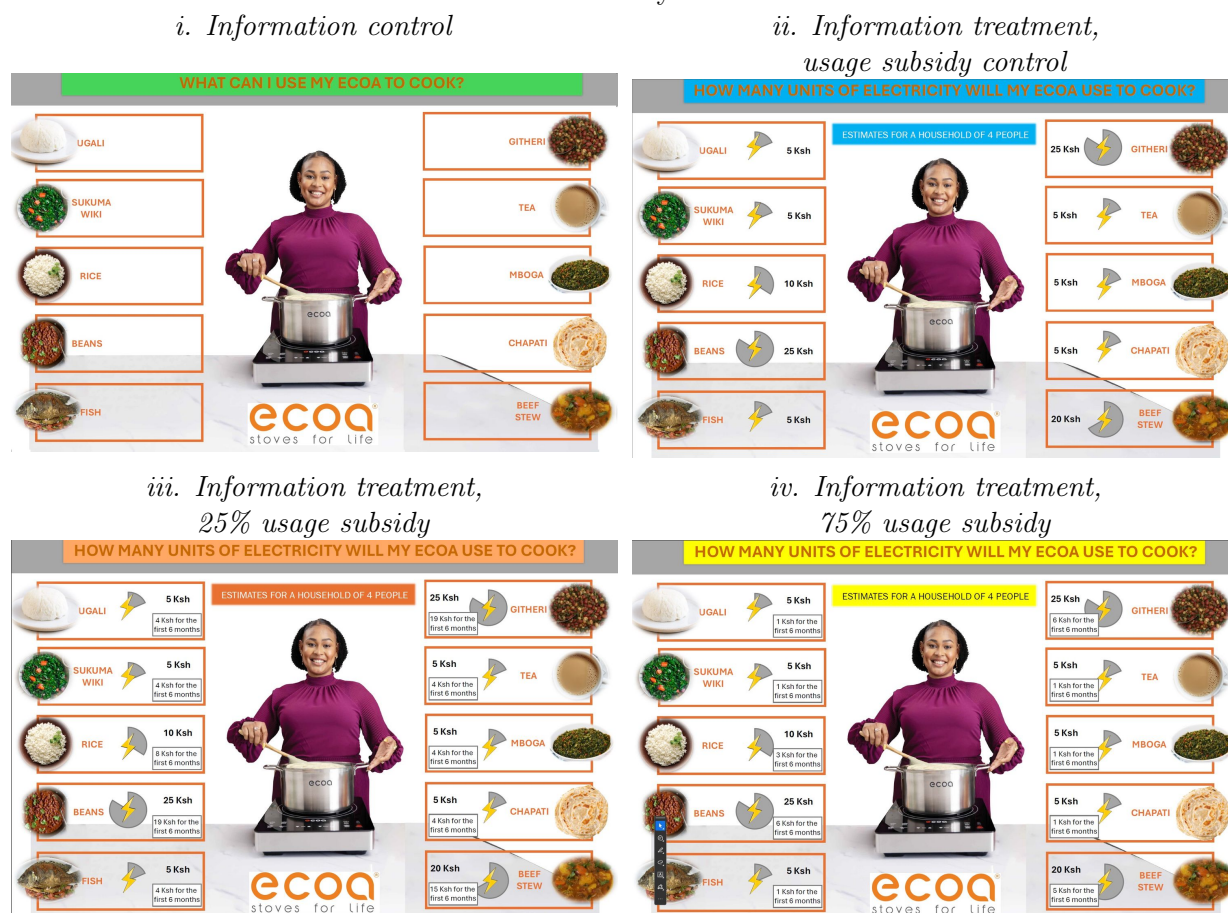
Notes: Random assignment for the 2,134 study participants. To preserve incentive compatibility, 4% of the sample received a different subsidy between 13% and 76%. [Section 3.4](#) discusses the randomized treatments in more detail.

Figure A3: Randomized information sheets

A. Charcoal sheet

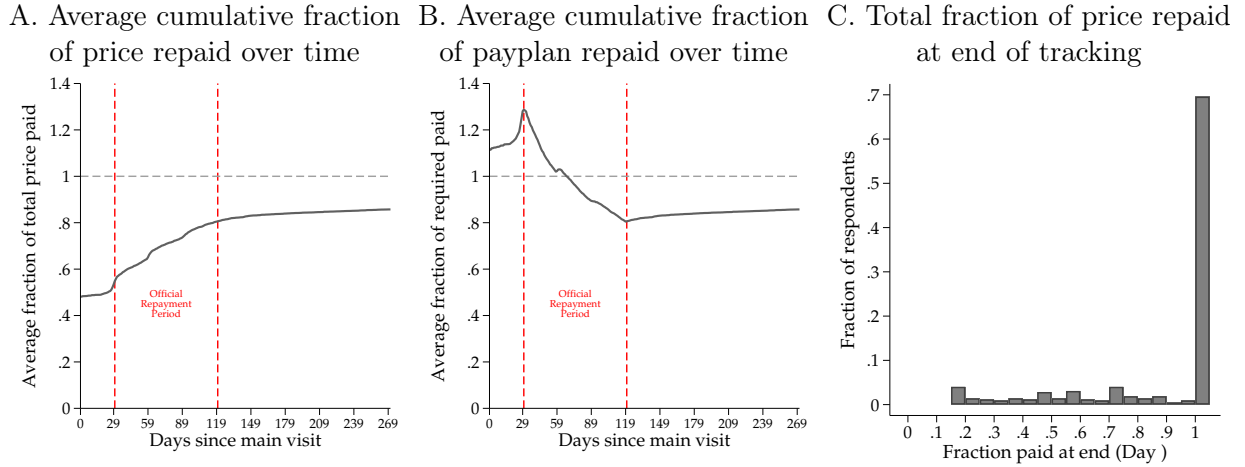


B. Electricity sheet



Notes: Each participant received one charcoal sheet and one electricity sheet. 14% of participants were in the information control group and received sheets with no cooking cost information (sheet *i* in both panels). 86% of participants were in the information treatment group and received cost information; participants in the usage subsidy and information treatment groups saw cooking costs that incorporated the price reduction (sheets *iii* and *iv* in Panel B).

Figure A4: Loan repayment patterns



Notes: Repayment data for 271 days since the main visit, the longest common follow-up horizon across all respondents. The US\$12 down payment required at day 0 constituted 16% of the price for those in the high price (low subsidy) group and 57% of the price for those in the low price (high subsidy) group. Panel A displays the average ratio of the amount respondents had paid to their total randomly assigned price on each day. In our study, borrowers have a 30-day grace period before the start of the official repayment period, which we mark as days 30–120 in the figure (dashed red lines). Panel B displays the average ratio of the amount respondents had paid to the amount cumulatively due on each day; values above 1 indicate respondents who were ahead of the smooth repayment schedule. Panel C displays a histogram of the fraction repaid by the 421 respondents who bought the electric stove with a loan.

II Appendix tables

Table A1: Summary statistics from respondent surveys at baseline

	Mean	SD	25 th	50 th	75 th	N
Number of residents	5.0	1.7	4	5	6	2134
Age	35.8	10.1	28	35	42	2134
Male (=1)	0.1	0.3	0	0	0	2134
Charcoal cooking time (minutes/day)	186.0	71.1	140	180	220	2134
Daily charcoal usage (KG per day)	3.1	1.0	2	3	4	2134
Number of charcoal stoves	1.1	0.3	1	1	1	2134
Charcoal spending (USD/month)	19.4	6.2	14	16	23	2134
Annual emissions from charcoal usage (tCO _{2e})	4.9	1.5	3	4	6	2134
Number of LPG stoves	0.6	0.6	0	1	1	2126
LPG spending (USD/month)	9.1	12.0	8	8	9	1228
Annual emissions from LPG usage (tCO _{2e})	0.3	0.4	0	0	0	1228
Number of wood stoves	0.1	0.4	0	0	0	2134
Wood spending (USD/month)	4.3	3.1	2	3	6	301
Respondent income (USD/month)	101.5	115.3	46	69	116	2127
Household income (USD/month)	246.9	204.8	116	196	300	1815
Monthly rent paid by household (30d; USD)	24.4	16.6	15	23	31	1797
Max willing to pay if phone broke (USD)	32.9	38.9	8	12	46	2134

Notes: Mean, standard deviation, and 25th, 50th, and 75th percentiles for key socioeconomic variables collected during the baseline survey. 31% of participants who use wood report getting some (or all) of their monthly wood consumption for free; rather than noisily estimate emissions, we omit CO_{2e} emissions from direct firewood burning from our calculations (if anything, this likely underestimates total CO_{2e} abatement from electric stove adoption and understates aggregate subsidy efficacy).

Table A2: Effect of induction stove adoption on charcoal stove usage

	Ordinary Least Squares		Instrumental Variables				Joint cooking (IV)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7) Charcoal	(8) Electric	(9) Net
Pre-period difference	0.06 (0.08)		-0.10 (0.17)		-0.10 (0.17)				
Post-period difference	-0.40*** (0.07)		-0.50*** (0.16)		-0.50*** (0.16)		-0.48*** (0.15)	0.59*** (0.03)	0.08 (0.15)
Pre-post difference		-0.48*** (0.06)		-0.30** (0.13)		-0.30** (0.13)			
Observations	223814	223814	5370250	5370250	223814	223814	204859	217234	204859
Respondents	2009	2009	2009	2009	2009	2009	1984	2080	1984
Control Mean	1.73	1.73	1.52	1.52	1.52	1.52	1.46	0.01	1.47
Level of data	Daily	Daily	Hourly	Hourly	Daily	Daily	Daily	Daily	Daily
Respondent FE?	No	Yes	No	Yes	No	Yes	No	No	No

Notes: Columns 1, 3, and 5 estimate the difference between adopters and non-adopters in the pre period (a baseline balance test) and in the post period (a treatment effect estimate). Columns 2, 4, and 6 estimate the adopter-nonadopter difference in the post period relative to the adopter-nonadopter difference in the pre period, where the latter is absorbed by respondent fixed effects. Columns 7, 8, and 9 evaluate post period data only. All regressions cluster standard errors by respondent and include date, days-since-visit, area, and field officer fixed effects. Columns 3 and 4 also include hour-of-day fixed effects. Columns 2, 4, and 6 include respondent fixed effects; the rest include the same controls as [Table 2](#). Column 3 uses hourly charcoal stove usage data and most closely corresponds to the pre period and post period treatment effects shown in [Figure 4](#); it replicates the specification reported in column 1 of [Table 2](#). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Alternative assumptions

Assumption	Subsidy cost per tCO _{2e}
Benchmark	US\$11.8
(1) One-year durability	US\$25.7
(2) Three-year durability	US\$8.6
(3) Fraction non-renewable biomass (fNRB) of 10%	US\$13.6
(4) Fraction non-renewable biomass (fNRB) of 100%	US\$9.0
(5) 100% fossil fuel generation (1 tCO _{2e} per MWh)	US\$12.3

Notes: Subsidy efficacy results under alternative assumptions. The benchmark is this paper's main estimate. Scenarios 1 and 2 assume that the stove's expected lifespan is one year or three years (the main estimate uses 2.2 years). Scenarios 3 and 4 assume that the fraction of the wood used for charcoal production that is non-renewable is either 10% or 100% (the main estimate assumes 38%). Scenario 5 assumes that Kenya's grid is powered entirely by coal, at 1 ton of CO_{2e} per MWh (the main estimate uses Kenya's grid average of 154 kg of CO_{2e} per MWh).

Table A4: Effect of credit and usage subsidy on willingness-to-pay

	(1)	(2)	(3)	(4)
Credit treatment	14.1*** (2.1)	14.7*** (1.9)	14.1*** (2.1)	14.7*** (1.9)
Subsidy	-2.0 (1.3)	-1.5 (1.2)		
Subsidy=25			-2.2 (1.5)	-0.8 (1.5)
Subsidy=75			-1.7 (1.5)	-2.3 (1.4)
Credit X Subsidy (any)	2.4 (2.6)	2.2 (2.4)		
Credit X 25% Subsidy			0.8 (3.0)	0.3 (2.8)
Credit X 75% Subsidy			4.0 (2.9)	4.0 (2.7)
Observations	2134	2134	2134	2134
Control Mean	19.5	19.5	19.5	19.5
SES controls	No	Yes	No	Yes

Notes: The effect of paying with credit, the usage subsidy treatment, and their interaction on willingness-to-pay. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Extensive and intensive margin impacts of usage subsidy

	Induction stove usage						
	Willingness-to-pay			Daily			Average
	(1) USD	(2) USD	(3) USD	(4) Wh	(5) Wh	(6) Log(Wh)	(7) Log(Wh)
Any subsidy	-0.41 (1.22)			20.24 (37.38)			
=25		-0.51 (1.44)			1.58 (43.80)		
=75		-0.31 (1.39)			40.51 (44.34)		
Subsidy value (USD)			-0.02 (0.10)				
Log(Marginal price)						-0.05 (0.05)	-0.07 (0.12)
Observations	2134	2134	2134	78505	78505	39714	612
Respondents				625	625	612	612
Control mean	28	28	28	496	496		

Notes: Results from ordinary least squares regressions estimating how the usage subsidy affects outcomes. Columns 1 and 4 pool the two subsidy levels of 25% and 75%. All columns control for randomization strata and other treatments. Columns 1–3 use one observation per respondent with heteroskedasticity-robust standard errors in parentheses. Columns 4–6 use data at the respondent-by-day level with date fixed effects and standard errors in parentheses clustered by respondent. Column 7 shows results when averaging daily usage to the respondent level. Stove usage data is missing for one of the 626 adopters.

Table A6: Impact of credit on daily induction stove usage

	All buyers		Marginal buyers		Low price buyers		First 30 days	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Daily	Respondent	Daily	Respondent	Daily	Respondent	Daily	Respondent
Credit (=1)	-122.6*** (37.4)	-123.4*** (39.7)	-92.6* (47.7)	-98.0* (53.2)	-98.6** (41.1)	-103.6** (44.3)	-137.6** (57.5)	-147.9** (66.1)
Observations	78505	625	44658	358	58792	470	9597	358
Respondents	625		358	358	470	470	358	358
Control Mean	578	574	561	554	561	554	621	624

Notes: Results from ordinary least squares regressions estimating how credit constraints affect the composition of marginal adopters. Columns 1 and 3 use respondent-by-day data with date fixed effects and standard errors clustered by respondent; columns 2 and 4 collapse usage to a respondent-level daily average and report heteroskedasticity-robust standard errors. All columns include the same controls as Table 4. Columns 5 and 6 confirm that these results are not driven by a mechanical device-lock effect. Since explicitly selecting on repayment introduces selection bias (we do not observe counterfactual repayment rates among those in the credit control group), we restrict the sample to adopters with a low price, among whom device lock-out occurred on only 18% of days (Online Appendix Figure B3 compares the groups). Columns 7 and 8 confirm that this effect holds in the first 30 days after adoption, when participants paying with cash would have just had a larger liquidity shock than participants paying with credit. Stove usage data is missing for one of the 626 adopters. Observations are winsorized at the 99th percentile. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.